In [4]:
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

In [5]:
```
df=pd.read_csv("customers.csv")
```

In [6]:
```
df
```

Out[6]:

| | ID | Year_Birth | Education | Marital_Status | Income | Kidhome | Teenhome | Dt_Customer | Recency | MntWines | ... | MntFishProducts | MntSweetProducts | MntGoldProds | NumDealsPurchases | NumWebPurchases | NumCatalogPurchases | NumStorePurchases | NumWebVisitsMonth | Complain | Response |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 5524 | 1957 | Graduation | Single | 58138.0 | 0 | 0 | 04-09-2012 | 58 | 635 | ... | 172 | 88 | 88 | 3 | 8 | 10 | 4 | 7 | 0 | 1 |
| **1** | 2174 | 1954 | Graduation | Single | 46344.0 | 1 | 1 | 08-03-2014 | 38 | 11 | ... | 2 | 1 | 6 | 2 | 1 | 1 | 2 | 5 | 0 | 0 |
| **2** | 4141 | 1965 | Graduation | Together | 71613.0 | 0 | 0 | 21-08-2013 | 26 | 426 | ... | 111 | 21 | 42 | 1 | 8 | 2 | 10 | 4 | 0 | 0 |
| **3** | 6182 | 1984 | Graduation | Together | 26646.0 | 1 | 0 | 10-02-2014 | 26 | 11 | ... | 10 | 3 | 5 | 2 | 2 | 0 | 4 | 6 | 0 | 0 |
| **4** | 5324 | 1981 | PhD | Married | 58293.0 | 1 | 0 | 19-01-2014 | 94 | 173 | ... | 46 | 27 | 15 | 5 | 5 | 3 | 6 | 5 | 0 | 0 |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| **2235** | 10870 | 1967 | Graduation | Married | 61223.0 | 0 | 1 | 13-06-2013 | 46 | 709 | ... | 42 | 118 | 247 | 2 | 9 | 3 | 4 | 5 | 0 | 0 |
| **2236** | 4001 | 1946 | PhD | Together | 64014.0 | 2 | 1 | 10-06-2014 | 56 | 406 | ... | 0 | 0 | 8 | 7 | 8 | 2 | 5 | 7 | 0 | 0 |
| **2237** | 7270 | 1981 | Graduation | Divorced | 56981.0 | 0 | 0 | 25-01-2014 | 91 | 908 | ... | 32 | 12 | 24 | 1 | 2 | 3 | 13 | 6 | 0 | 0 |
| **2238** | 8235 | 1956 | Master | Together | 69245.0 | 0 | 1 | 24-01-2014 | 8 | 428 | ... | 80 | 30 | 61 | 2 | 6 | 5 | 10 | 3 | 0 | 0 |
| **2239** | 9405 | 1954 | PhD | Married | 52869.0 | 1 | 1 | 15-10-2012 | 40 | 84 | ... | 2 | 1 | 21 | 3 | 3 | 1 | 4 | 7 | 0 | 1 |

2240 rows × 22 columns

In [7]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2240 entries, 0 to 2239
Data columns (total 22 columns):
 #   Column               Non-Null Count  Dtype  
---  ------               --------------  -----  
 0   ID                   2240 non-null   int64  
 1   Year_Birth           2240 non-null   int64  
 2   Education            2240 non-null   object 
 3   Marital_Status       2240 non-null   object 
 4   Income               2216 non-null   float64
 5   Kidhome              2240 non-null   int64  
 6   Teenhome             2240 non-null   int64  
 7   Dt_Customer          2240 non-null   object 
 8   Recency              2240 non-null   int64  
 9   MntWines             2240 non-null   int64  
 10  MntFruits            2240 non-null   int64  
 11  MntMeatProducts      2240 non-null   int64  
 12  MntFishProducts      2240 non-null   int64  
 13  MntSweetProducts     2240 non-null   int64  
 14  MntGoldProds         2240 non-null   int64  
 15  NumDealsPurchases    2240 non-null   int64  
 16  NumWebPurchases      2240 non-null   int64  
 17  NumCatalogPurchases  2240 non-null   int64  
 18  NumStorePurchases    2240 non-null   int64  
 19  NumWebVisitsMonth    2240 non-null   int64  
 20  Complain             2240 non-null   int64  
 21  Response             2240 non-null   int64  
dtypes: float64(1), int64(18), object(3)
memory usage: 385.1+ KB
```

In [8]:
```
df.isnull().sum()
```

Out[8]:
```
ID                      0
Year_Birth              0
Education               0
Marital_Status          0
Income                 24
Kidhome                 0
Teenhome                0
Dt_Customer             0
Recency                 0
MntWines                0
MntFruits               0
MntMeatProducts         0
MntFishProducts         0
MntSweetProducts        0
MntGoldProds            0
NumDealsPurchases       0
NumWebPurchases         0
NumCatalogPurchases     0
NumStorePurchases       0
NumWebVisitsMonth       0
Complain                0
Response                0
dtype: int64
```

In [9]:
```
df.shape
```

Out[9]:
```
(2240, 22)
```

## Handle Missing Data Values

In [10]:
```
#Data Preprocessing
from sklearn.impute import SimpleImputer
imputer=SimpleImputer(strategy="median")
df["Income"]=imputer.fit_transform(df[["Income"]]).ravel()
```

In [11]:
```
df
```

Out[11]:

| | ID | Year_Birth | Education | Marital_Status | Income | Kidhome | Teenhome | Dt_Customer | Recency | MntWines | ... | MntFishProducts | MntSweetProducts | MntGoldProds | NumDealsPurchases | NumWebPurchases | NumCatalogPurchases | NumStorePurchases | NumWebVisitsMonth | Complain | Response |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 5524 | 1957 | Graduation | Single | 58138.0 | 0 | 0 | 04-09-2012 | 58 | 635 | ... | 172 | 88 | 88 | 3 | 8 | 10 | 4 | 7 | 0 | 1 |
| **1** | 2174 | 1954 | Graduation | Single | 46344.0 | 1 | 1 | 08-03-2014 | 38 | 11 | ... | 2 | 1 | 6 | 2 | 1 | 1 | 2 | 5 | 0 | 0 |
| **2** | 4141 | 1965 | Graduation | Together | 71613.0 | 0 | 0 | 21-08-2013 | 26 | 426 | ... | 111 | 21 | 42 | 1 | 8 | 2 | 10 | 4 | 0 | 0 |
| **3** | 6182 | 1984 | Graduation | Together | 26646.0 | 1 | 0 | 10-02-2014 | 26 | 11 | ... | 10 | 3 | 5 | 2 | 2 | 0 | 4 | 6 | 0 | 0 |
| **4** | 5324 | 1981 | PhD | Married | 58293.0 | 1 | 0 | 19-01-2014 | 94 | 173 | ... | 46 | 27 | 15 | 5 | 5 | 3 | 6 | 5 | 0 | 0 |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| **2235** | 10870 | 1967 | Graduation | Married | 61223.0 | 0 | 1 | 13-06-2013 | 46 | 709 | ... | 42 | 118 | 247 | 2 | 9 | 3 | 4 | 5 | 0 | 0 |
| **2236** | 4001 | 1946 | PhD | Together | 64014.0 | 2 | 1 | 10-06-2014 | 56 | 406 | ... | 0 | 0 | 8 | 7 | 8 | 2 | 5 | 7 | 0 | 0 |
| **2237** | 7270 | 1981 | Graduation | Divorced | 56981.0 | 0 | 0 | 25-01-2014 | 91 | 908 | ... | 32 | 12 | 24 | 1 | 2 | 3 | 13 | 6 | 0 | 0 |
| **2238** | 8235 | 1956 | Master | Together | 69245.0 | 0 | 1 | 24-01-2014 | 8 | 428 | ... | 80 | 30 | 61 | 2 | 6 | 5 | 10 | 3 | 0 | 0 |
| **2239** | 9405 | 1954 | PhD | Married | 52869.0 | 1 | 1 | 15-10-2012 | 40 | 84 | ... | 2 | 1 | 21 | 3 | 3 | 1 | 4 | 7 | 0 | 1 |

2240 rows × 22 columns

In [12]:
```
df.isnull().sum()
```

Out[12]:
```
ID                     0
Year_Birth             0
Education              0
Marital_Status         0
Income                 0
Kidhome                0
Teenhome               0
Dt_Customer            0
Recency                0
MntWines               0
MntFruits              0
MntMeatProducts        0
MntFishProducts        0
MntSweetProducts       0
MntGoldProds           0
NumDealsPurchases      0
NumWebPurchases        0
NumCatalogPurchases    0
NumStorePurchases      0
NumWebVisitsMonth      0
Complain               0
Response               0
dtype: int64
```

## Feature Engineering

In [13]:
```
df["Age"]=2026-df["Year_Birth"]
```

In [14]:
```
df
```

Out[14]:

| | ID | Year_Birth | Education | Marital_Status | Income | Kidhome | Teenhome | Dt_Customer | Recency | MntWines | ... | MntSweetProducts | MntGoldProds | NumDealsPurchases | NumWebPurchases | NumCatalogPurchases | NumStorePurchases | NumWebVisitsMonth | Complain | Response | Age |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 5524 | 1957 | Graduation | Single | 58138.0 | 0 | 0 | 04-09-2012 | 58 | 635 | ... | 88 | 88 | 3 | 8 | 10 | 4 | 7 | 0 | 1 | 69 |
| **1** | 2174 | 1954 | Graduation | Single | 46344.0 | 1 | 1 | 08-03-2014 | 38 | 11 | ... | 1 | 6 | 2 | 1 | 1 | 2 | 5 | 0 | 0 | 72 |
| **2** | 4141 | 1965 | Graduation | Together | 71613.0 | 0 | 0 | 21-08-2013 | 26 | 426 | ... | 21 | 42 | 1 | 8 | 2 | 10 | 4 | 0 | 0 | 61 |
| **3** | 6182 | 1984 | Graduation | Together | 26646.0 | 1 | 0 | 10-02-2014 | 26 | 11 | ... | 3 | 5 | 2 | 2 | 0 | 4 | 6 | 0 | 0 | 42 |
| **4** | 5324 | 1981 | PhD | Married | 58293.0 | 1 | 0 | 19-01-2014 | 94 | 173 | ... | 27 | 15 | 5 | 5 | 3 | 6 | 5 | 0 | 0 | 45 |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| **2235** | 10870 | 1967 | Graduation | Married | 61223.0 | 0 | 1 | 13-06-2013 | 46 | 709 | ... | 118 | 247 | 2 | 9 | 3 | 4 | 5 | 0 | 0 | 59 |
| **2236** | 4001 | 1946 | PhD | Together | 64014.0 | 2 | 1 | 10-06-2014 | 56 | 406 | ... | 0 | 8 | 7 | 8 | 2 | 5 | 7 | 0 | 0 | 80 |
| **2237** | 7270 | 1981 | Graduation | Divorced | 56981.0 | 0 | 0 | 25-01-2014 | 91 | 908 | ... | 12 | 24 | 1 | 2 | 3 | 13 | 6 | 0 | 0 | 45 |
| **2238** | 8235 | 1956 | Master | Together | 69245.0 | 0 | 1 | 24-01-2014 | 8 | 428 | ... | 30 | 61 | 2 | 6 | 5 | 10 | 3 | 0 | 0 | 70 |
| **2239** | 9405 | 1954 | PhD | Married | 52869.0 | 1 | 1 | 15-10-2012 | 40 | 84 | ... | 1 | 21 | 3 | 3 | 1 | 4 | 7 | 0 | 1 | 72 |

2240 rows × 23 columns

In [15]:
```
df.head(20)
```

Out[15]:

| | ID | Year_Birth | Education | Marital_Status | Income | Kidhome | Teenhome | Dt_Customer | Recency | MntWines | ... | MntSweetProducts | MntGoldProds | NumDealsPurchases | NumWebPurchases | NumCatalogPurchases | NumStorePurchases | NumWebVisitsMonth | Complain | Response | Age |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 5524 | 1957 | Graduation | Single | 58138.0 | 0 | 0 | 04-09-2012 | 58 | 635 | ... | 88 | 88 | 3 | 8 | 10 | 4 | 7 | 0 | 1 | 69 |
| **1** | 2174 | 1954 | Graduation | Single | 46344.0 | 1 | 1 | 08-03-2014 | 38 | 11 | ... | 1 | 6 | 2 | 1 | 1 | 2 | 5 | 0 | 0 | 72 |
| **2** | 4141 | 1965 | Graduation | Together | 71613.0 | 0 | 0 | 21-08-2013 | 26 | 426 | ... | 21 | 42 | 1 | 8 | 2 | 10 | 4 | 0 | 0 | 61 |
| **3** | 6182 | 1984 | Graduation | Together | 26646.0 | 1 | 0 | 10-02-2014 | 26 | 11 | ... | 3 | 5 | 2 | 2 | 0 | 4 | 6 | 0 | 0 | 42 |
| **4** | 5324 | 1981 | PhD | Married | 58293.0 | 1 | 0 | 19-01-2014 | 94 | 173 | ... | 27 | 15 | 5 | 5 | 3 | 6 | 5 | 0 | 0 | 45 |
| **5** | 7446 | 1967 | Master | Together | 62513.0 | 0 | 1 | 09-09-2013 | 16 | 520 | ... | 42 | 14 | 2 | 6 | 4 | 10 | 6 | 0 | 0 | 59 |
| **6** | 965 | 1971 | Graduation | Divorced | 55635.0 | 0 | 1 | 13-11-2012 | 34 | 235 | ... | 49 | 27 | 4 | 7 | 3 | 7 | 6 | 0 | 0 | 55 |
| **7** | 6177 | 1985 | PhD | Married | 33454.0 | 1 | 0 | 08-05-2013 | 32 | 76 | ... | 1 | 23 | 2 | 4 | 0 | 4 | 8 | 0 | 0 | 41 |
| **8** | 4855 | 1974 | PhD | Together | 30351.0 | 1 | 0 | 06-06-2013 | 19 | 14 | ... | 3 | 2 | 1 | 3 | 0 | 2 | 9 | 0 | 1 | 52 |
| **9** | 5899 | 1950 | PhD | Together | 5648.0 | 1 | 1 | 13-03-2014 | 68 | 28 | ... | 1 | 13 | 1 | 1 | 0 | 0 | 20 | 0 | 0 | 76 |
| **10** | 1994 | 1983 | Graduation | Married | 51381.5 | 1 | 0 | 15-11-2013 | 11 | 5 | ... | 2 | 1 | 1 | 1 | 0 | 2 | 7 | 0 | 0 | 43 |
| **11** | 387 | 1976 | Basic | Married | 7500.0 | 0 | 0 | 13-11-2012 | 59 | 6 | ... | 1 | 16 | 1 | 2 | 0 | 3 | 8 | 0 | 0 | 50 |
| **12** | 2125 | 1959 | Graduation | Divorced | 63033.0 | 0 | 0 | 15-11-2013 | 82 | 194 | ... | 112 | 30 | 1 | 3 | 4 | 8 | 2 | 0 | 0 | 67 |
| **13** | 8180 | 1952 | Master | Divorced | 59354.0 | 1 | 1 | 15-11-2013 | 53 | 233 | ... | 5 | 14 | 3 | 6 | 1 | 5 | 6 | 0 | 0 | 74 |
| **14** | 2569 | 1987 | Graduation | Married | 17323.0 | 0 | 0 | 10-10-2012 | 38 | 3 | ... | 1 | 5 | 1 | 1 | 0 | 3 | 8 | 0 | 0 | 39 |
| **15** | 2114 | 1946 | PhD | Single | 82800.0 | 0 | 0 | 24-11-2012 | 23 | 1006 | ... | 68 | 45 | 1 | 7 | 6 | 12 | 3 | 0 | 1 | 80 |
| **16** | 9736 | 1980 | Graduation | Married | 41850.0 | 1 | 1 | 24-12-2012 | 51 | 53 | ... | 13 | 4 | 3 | 3 | 0 | 3 | 8 | 0 | 0 | 46 |
| **17** | 4939 | 1946 | Graduation | Together | 37760.0 | 0 | 0 | 31-08-2012 | 20 | 84 | ... | 12 | 28 | 2 | 4 | 1 | 6 | 7 | 0 | 0 | 80 |
| **18** | 6565 | 1949 | Master | Married | 76995.0 | 0 | 1 | 28-03-2013 | 91 | 1012 | ... | 16 | 176 | 2 | 11 | 4 | 9 | 5 | 0 | 0 | 77 |
| **19** | 2278 | 1985 | 2n Cycle | Single | 33812.0 | 1 | 0 | 03-11-2012 | 86 | 4 | ... | 24 | 39 | 2 | 2 | 1 | 3 | 6 | 0 | 0 | 41 |

20 rows × 23 columns

In [16]:
```
df.describe()
```

Out[16]:

| | ID | Year_Birth | Income | Kidhome | Teenhome | Recency | MntWines | MntFruits | MntMeatProducts | MntFishProducts | MntSweetProducts | MntGoldProds | NumDealsPurchases | NumWebPurchases | NumCatalogPurchases | NumStorePurchases | NumWebVisitsMonth | Complain |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **count** | 2240.000000 | 2240.000000 | 2240.000000 | 2240.000000 | 2240.000000 | 2240.000000 | 2240.000000 | 2240.000000 | 2240.000000 | 2240.000000 | 2240.000000 | 2240.000000 | 2240.000000 | 2240.000000 | 2240.000000 | 2240.000000 | 2240.000000 | 2240.000000 |
| **mean** | 5592.159821 | 1968.805804 | 52237.975446 | 0.444196 | 0.506250 | 49.109375 | 303.935714 | 26.302232 | 166.950000 | 37.525446 | 27.062946 | 44.021875 | 2.325000 | 4.084821 | 2.662054 | 5.790179 | 5.316518 | 0.009375 |
| **std** | 3246.662198 | 11.984069 | 25037.955891 | 0.538398 | 0.544538 | 28.962453 | 336.597393 | 39.773434 | 225.715373 | 54.628979 | 41.280498 | 52.167439 | 1.932238 | 2.778714 | 2.923101 | 3.250958 | 2.426645 | 0.096391 |
| **min** | 0.000000 | 1893.000000 | 1730.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 |
| **25%** | 2828.250000 | 1959.000000 | 35538.750000 | 0.000000 | 0.000000 | 24.000000 | 23.750000 | 1.000000 | 16.000000 | 3.000000 | 1.000000 | 9.000000 | 1.000000 | 2.000000 | 0.000000 | 3.000000 | 3.000000 | 0.000000 |
| **50%** | 5458.500000 | 1970.000000 | 51381.500000 | 0.000000 | 0.000000 | 49.000000 | 173.500000 | 8.000000 | 67.000000 | 12.000000 | 8.000000 | 24.000000 | 2.000000 | 4.000000 | 2.000000 | 5.000000 | 6.000000 | 0.000000 |
| **75%** | 8427.750000 | 1977.000000 | 68289.750000 | 1.000000 | 1.000000 | 74.000000 | 504.250000 | 33.000000 | 232.000000 | 50.000000 | 33.000000 | 56.000000 | 3.000000 | 6.000000 | 4.000000 | 8.000000 | 7.000000 | 0.000000 |
| **max** | 11191.000000 | 1996.000000 | 666666.000000 | 2.000000 | 2.000000 | 99.000000 | 1493.000000 | 199.000000 | 1725.000000 | 259.000000 | 263.000000 | 362.000000 | 15.000000 | 27.000000 | 28.000000 | 13.000000 | 20.000000 | 1.000000 |

In [17]:
```
df["Dt_Customer"]=pd.to_datetime(df["Dt_Customer"],dayfirst=True)
reference_date=df["Dt_Customer"].max()
df["Customer_Tenure_Days"]=(reference_date-df["Dt_Customer"]).dt.days
```

In [18]:
```
df
```

Out[18]:

| | ID | Year_Birth | Education | Marital_Status | Income | Kidhome | Teenhome | Dt_Customer | Recency | MntWines | ... | MntGoldProds | NumDealsPurchases | NumWebPurchases | NumCatalogPurchases | NumStorePurchases | NumWebVisitsMonth | Complain | Response | Age | Customer_Tenure_Days |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 5524 | 1957 | Graduation | Single | 58138.0 | 0 | 0 | 2012-09-04 | 58 | 635 | ... | 88 | 3 | 8 | 10 | 4 | 7 | 0 | 1 | 69 | 663 |
| **1** | 2174 | 1954 | Graduation | Single | 46344.0 | 1 | 1 | 2014-03-08 | 38 | 11 | ... | 6 | 2 | 1 | 1 | 2 | 5 | 0 | 0 | 72 | 113 |
| **2** | 4141 | 1965 | Graduation | Together | 71613.0 | 0 | 0 | 2013-08-21 | 26 | 426 | ... | 42 | 1 | 8 | 2 | 10 | 4 | 0 | 0 | 61 | 312 |
| **3** | 6182 | 1984 | Graduation | Together | 26646.0 | 1 | 0 | 2014-02-10 | 26 | 11 | ... | 5 | 2 | 2 | 0 | 4 | 6 | 0 | 0 | 42 | 139 |
| **4** | 5324 | 1981 | PhD | Married | 58293.0 | 1 | 0 | 2014-01-19 | 94 | 173 | ... | 15 | 5 | 5 | 3 | 6 | 5 | 0 | 0 | 45 | 161 |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| **2235** | 10870 | 1967 | Graduation | Married | 61223.0 | 0 | 1 | 2013-06-13 | 46 | 709 | ... | 247 | 2 | 9 | 3 | 4 | 5 | 0 | 0 | 59 | 381 |
| **2236** | 4001 | 1946 | PhD | Together | 64014.0 | 2 | 1 | 2014-06-10 | 56 | 406 | ... | 8 | 7 | 8 | 2 | 5 | 7 | 0 | 0 | 80 | 19 |
| **2237** | 7270 | 1981 | Graduation | Divorced | 56981.0 | 0 | 0 | 2014-01-25 | 91 | 908 | ... | 24 | 1 | 2 | 3 | 13 | 6 | 0 | 0 | 45 | 155 |
| **2238** | 8235 | 1956 | Master | Together | 69245.0 | 0 | 1 | 2014-01-24 | 8 | 428 | ... | 61 | 2 | 6 | 5 | 10 | 3 | 0 | 0 | 70 | 156 |
| **2239** | 9405 | 1954 | PhD | Married | 52869.0 | 1 | 1 | 2012-10-15 | 40 | 84 | ... | 21 | 3 | 3 | 1 | 4 | 7 | 0 | 1 | 72 | 622 |

2240 rows × 24 columns

In [19]:
```
print(df["Dt_Customer"].dtype)
```

```
datetime64[ns]
```

In [20]:
```
df["Dt_Customer"].max()
```

Out[20]:
```
Timestamp('2014-06-29 00:00:00')
```

In [21]:
```
df.columns
```

Out[21]:
```
Index(['ID', 'Year_Birth', 'Education', 'Marital_Status', 'Income', 'Kidhome',
       'Teenhome', 'Dt_Customer', 'Recency', 'MntWines', 'MntFruits',
       'MntMeatProducts', 'MntFishProducts', 'MntSweetProducts',
       'MntGoldProds', 'NumDealsPurchases', 'NumWebPurchases',
       'NumCatalogPurchases', 'NumStorePurchases', 'NumWebVisitsMonth',
       'Complain', 'Response', 'Age', 'Customer_Tenure_Days'],
      dtype='object')
```

In [22]:
```
df["Total_Spending"]=df["MntWines"]+df["MntFruits"]+df["MntMeatProducts"]+df["MntFishProducts"]+df["MntSweetProducts"]+df["MntGoldProds"]
```

In [23]:
```
df
```

Out[23]:

| | ID | Year_Birth | Education | Marital_Status | Income | Kidhome | Teenhome | Dt_Customer | Recency | MntWines | ... | NumDealsPurchases | NumWebPurchases | NumCatalogPurchases | NumStorePurchases | NumWebVisitsMonth | Complain | Response | Age | Customer_Tenure_Days | Total_Spending |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 5524 | 1957 | Graduation | Single | 58138.0 | 0 | 0 | 2012-09-04 | 58 | 635 | ... | 3 | 8 | 10 | 4 | 7 | 0 | 1 | 69 | 663 | 1617 |
| **1** | 2174 | 1954 | Graduation | Single | 46344.0 | 1 | 1 | 2014-03-08 | 38 | 11 | ... | 2 | 1 | 1 | 2 | 5 | 0 | 0 | 72 | 113 | 27 |
| **2** | 4141 | 1965 | Graduation | Together | 71613.0 | 0 | 0 | 2013-08-21 | 26 | 426 | ... | 1 | 8 | 2 | 10 | 4 | 0 | 0 | 61 | 312 | 776 |
| **3** | 6182 | 1984 | Graduation | Together | 26646.0 | 1 | 0 | 2014-02-10 | 26 | 11 | ... | 2 | 2 | 0 | 4 | 6 | 0 | 0 | 42 | 139 | 53 |
| **4** | 5324 | 1981 | PhD | Married | 58293.0 | 1 | 0 | 2014-01-19 | 94 | 173 | ... | 5 | 5 | 3 | 6 | 5 | 0 | 0 | 45 | 161 | 422 |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| **2235** | 10870 | 1967 | Graduation | Married | 61223.0 | 0 | 1 | 2013-06-13 | 46 | 709 | ... | 2 | 9 | 3 | 4 | 5 | 0 | 0 | 59 | 381 | 1341 |
| **2236** | 4001 | 1946 | PhD | Together | 64014.0 | 2 | 1 | 2014-06-10 | 56 | 406 | ... | 7 | 8 | 2 | 5 | 7 | 0 | 0 | 80 | 19 | 444 |
| **2237** | 7270 | 1981 | Graduation | Divorced | 56981.0 | 0 | 0 | 2014-01-25 | 91 | 908 | ... | 1 | 2 | 3 | 13 | 6 | 0 | 0 | 45 | 155 | 1241 |
| **2238** | 8235 | 1956 | Master | Together | 69245.0 | 0 | 1 | 2014-01-24 | 8 | 428 | ... | 2 | 6 | 5 | 10 | 3 | 0 | 0 | 70 | 156 | 843 |
| **2239** | 9405 | 1954 | PhD | Married | 52869.0 | 1 | 1 | 2012-10-15 | 40 | 84 | ... | 3 | 3 | 1 | 4 | 7 | 0 | 1 | 72 | 622 | 172 |

2240 rows × 25 columns

In [24]:
```
df["Total_children"]=df["Kidhome"]+df["Teenhome"]
```

In [25]:
```
df
```

Out[25]:

| | ID | Year_Birth | Education | Marital_Status | Income | Kidhome | Teenhome | Dt_Customer | Recency | MntWines | ... | NumWebPurchases | NumCatalogPurchases | NumStorePurchases | NumWebVisitsMonth | Complain | Response | Age | Customer_Tenure_Days | Total_Spending | Total_children |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 5524 | 1957 | Graduation | Single | 58138.0 | 0 | 0 | 2012-09-04 | 58 | 635 | ... | 8 | 10 | 4 | 7 | 0 | 1 | 69 | 663 | 1617 | 0 |
| **1** | 2174 | 1954 | Graduation | Single | 46344.0 | 1 | 1 | 2014-03-08 | 38 | 11 | ... | 1 | 1 | 2 | 5 | 0 | 0 | 72 | 113 | 27 | 2 |
| **2** | 4141 | 1965 | Graduation | Together | 71613.0 | 0 | 0 | 2013-08-21 | 26 | 426 | ... | 8 | 2 | 10 | 4 | 0 | 0 | 61 | 312 | 776 | 0 |
| **3** | 6182 | 1984 | Graduation | Together | 26646.0 | 1 | 0 | 2014-02-10 | 26 | 11 | ... | 2 | 0 | 4 | 6 | 0 | 0 | 42 | 139 | 53 | 1 |
| **4** | 5324 | 1981 | PhD | Married | 58293.0 | 1 | 0 | 2014-01-19 | 94 | 173 | ... | 5 | 3 | 6 | 5 | 0 | 0 | 45 | 161 | 422 | 1 |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| **2235** | 10870 | 1967 | Graduation | Married | 61223.0 | 0 | 1 | 2013-06-13 | 46 | 709 | ... | 9 | 3 | 4 | 5 | 0 | 0 | 59 | 381 | 1341 | 1 |
| **2236** | 4001 | 1946 | PhD | Together | 64014.0 | 2 | 1 | 2014-06-10 | 56 | 406 | ... | 8 | 2 | 5 | 7 | 0 | 0 | 80 | 19 | 444 | 3 |
| **2237** | 7270 | 1981 | Graduation | Divorced | 56981.0 | 0 | 0 | 2014-01-25 | 91 | 908 | ... | 2 | 3 | 13 | 6 | 0 | 0 | 45 | 155 | 1241 | 0 |
| **2238** | 8235 | 1956 | Master | Together | 69245.0 | 0 | 1 | 2014-01-24 | 8 | 428 | ... | 6 | 5 | 10 | 3 | 0 | 0 | 70 | 156 | 843 | 1 |
| **2239** | 9405 | 1954 | PhD | Married | 52869.0 | 1 | 1 | 2012-10-15 | 40 | 84 | ... | 3 | 1 | 4 | 7 | 0 | 1 | 72 | 622 | 172 | 2 |

2240 rows × 26 columns

In [26]:
```
df.columns
```

Out[26]:
```
Index(['ID', 'Year_Birth', 'Education', 'Marital_Status', 'Income', 'Kidhome',
       'Teenhome', 'Dt_Customer', 'Recency', 'MntWines', 'MntFruits',
       'MntMeatProducts', 'MntFishProducts', 'MntSweetProducts',
       'MntGoldProds', 'NumDealsPurchases', 'NumWebPurchases',
       'NumCatalogPurchases', 'NumStorePurchases', 'NumWebVisitsMonth',
       'Complain', 'Response', 'Age', 'Customer_Tenure_Days', 'Total_Spending',
       'Total_children'],
      dtype='object')
```

In [27]:
```
df["Total_children"].max()
```

Out[27]:
```
3
```

In [28]:
```
df["Education"].value_counts()
```

Out[28]:
```
Education
Graduation    1127
PhD            486
Master         370
2n Cycle       203
Basic           54
Name: count, dtype: int64
```

In [29]:
```
df["Education"]=df["Education"].replace({
    "Basic":"Undergraduate",
    "Graduation":"Graduate",
    "Master":"Postgraduate",
    "PhD":"Postgraduate",
    "2n Cycle":"Undergraduate"
})
```

In [30]:
```
df["Education"].value_counts()
```

Out[30]:
```
Education
Graduate         1127
Postgraduate      856
Undergraduate     257
Name: count, dtype: int64
```

In [31]:
```
df["Marital_Status"].value_counts()
```

Out[31]:
```
Marital_Status
Married     864
Together    580
Single      480
Divorced    232
Widow        77
Alone         3
Absurd        2
YOLO          2
Name: count, dtype: int64
```

In [32]:
```
df["Living_with"]=df["Marital_Status"].replace({"Married":"Partner","Together":"Partner","Single":"Alone","Divorced":"Alone","Widow":"Alone","Absurd":"Alone","YOLO":"Alone"})
```

In [33]:
```
df["Living_with"]
```

Out[33]:
```
0         Alone
1         Alone
2       Partner
3       Partner
4       Partner
         ...   
2235    Partner
2236    Partner
2237      Alone
2238    Partner
2239    Partner
Name: Living_with, Length: 2240, dtype: object
```

In [34]:
```
df
```

Out[34]:

| | ID | Year_Birth | Education | Marital_Status | Income | Kidhome | Teenhome | Dt_Customer | Recency | MntWines | ... | NumCatalogPurchases | NumStorePurchases | NumWebVisitsMonth | Complain | Response | Age | Customer_Tenure_Days | Total_Spending | Total_children | Living_with |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 5524 | 1957 | Graduate | Single | 58138.0 | 0 | 0 | 2012-09-04 | 58 | 635 | ... | 10 | 4 | 7 | 0 | 1 | 69 | 663 | 1617 | 0 | Alone |
| **1** | 2174 | 1954 | Graduate | Single | 46344.0 | 1 | 1 | 2014-03-08 | 38 | 11 | ... | 1 | 2 | 5 | 0 | 0 | 72 | 113 | 27 | 2 | Alone |
| **2** | 4141 | 1965 | Graduate | Together | 71613.0 | 0 | 0 | 2013-08-21 | 26 | 426 | ... | 2 | 10 | 4 | 0 | 0 | 61 | 312 | 776 | 0 | Partner |
| **3** | 6182 | 1984 | Graduate | Together | 26646.0 | 1 | 0 | 2014-02-10 | 26 | 11 | ... | 0 | 4 | 6 | 0 | 0 | 42 | 139 | 53 | 1 | Partner |
| **4** | 5324 | 1981 | Postgraduate | Married | 58293.0 | 1 | 0 | 2014-01-19 | 94 | 173 | ... | 3 | 6 | 5 | 0 | 0 | 45 | 161 | 422 | 1 | Partner |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| **2235** | 10870 | 1967 | Graduate | Married | 61223.0 | 0 | 1 | 2013-06-13 | 46 | 709 | ... | 3 | 4 | 5 | 0 | 0 | 59 | 381 | 1341 | 1 | Partner |
| **2236** | 4001 | 1946 | Postgraduate | Together | 64014.0 | 2 | 1 | 2014-06-10 | 56 | 406 | ... | 2 | 5 | 7 | 0 | 0 | 80 | 19 | 444 | 3 | Partner |
| **2237** | 7270 | 1981 | Graduate | Divorced | 56981.0 | 0 | 0 | 2014-01-25 | 91 | 908 | ... | 3 | 13 | 6 | 0 | 0 | 45 | 155 | 1241 | 0 | Alone |
| **2238** | 8235 | 1956 | Postgraduate | Together | 69245.0 | 0 | 1 | 2014-01-24 | 8 | 428 | ... | 5 | 10 | 3 | 0 | 0 | 70 | 156 | 843 | 1 | Partner |
| **2239** | 9405 | 1954 | Postgraduate | Married | 52869.0 | 1 | 1 | 2012-10-15 | 40 | 84 | ... | 1 | 4 | 7 | 0 | 1 | 72 | 622 | 172 | 2 | Partner |

2240 rows × 27 columns

In [35]:
```
df.columns
```

Out[35]:
```
Index(['ID', 'Year_Birth', 'Education', 'Marital_Status', 'Income', 'Kidhome',
       'Teenhome', 'Dt_Customer', 'Recency', 'MntWines', 'MntFruits',
       'MntMeatProducts', 'MntFishProducts', 'MntSweetProducts',
       'MntGoldProds', 'NumDealsPurchases', 'NumWebPurchases',
       'NumCatalogPurchases', 'NumStorePurchases', 'NumWebVisitsMonth',
       'Complain', 'Response', 'Age', 'Customer_Tenure_Days', 'Total_Spending',
       'Total_children', 'Living_with'],
      dtype='object')
```

In [36]:
```
cols=["ID","Year_Birth","Marital_Status","Kidhome","Teenhome","Dt_Customer"]
spending_cols=["MntWines","MntFruits","MntMeatProducts","MntFishProducts","MntSweetProducts","MntGoldProds"]
cols_to_drop=cols+spending_cols
df_cleaned=df.drop(columns=cols_to_drop)
```

In [37]:
```
cols_to_drop
```

Out[37]:
```
['ID',
 'Year_Birth',
 'Marital_Status',
 'Kidhome',
 'Teenhome',
 'Dt_Customer',
 'MntWines',
 'MntFruits',
 'MntMeatProducts',
 'MntFishProducts',
 'MntSweetProducts',
 'MntGoldProds']
```

In [38]:
```
df_cleaned
```

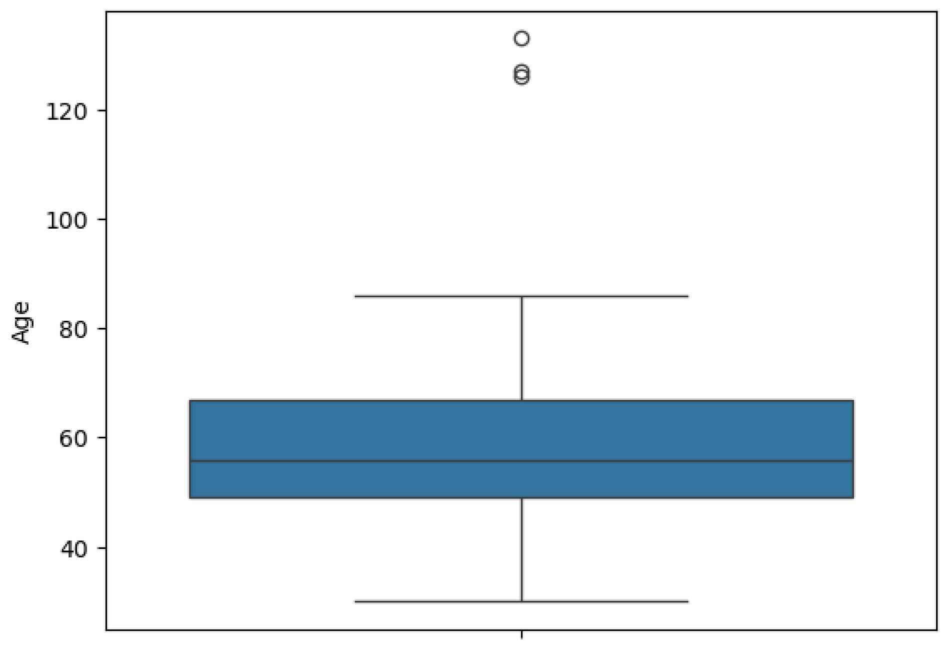
Out[38]:

| | Education | Income | Recency | NumDealsPurchases | NumWebPurchases | NumCatalogPurchases | NumStorePurchases | NumWebVisitsMonth | Complain | Response | Age | Customer_Tenure_Days | Total_Spending | Total_children | Living_with |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | Graduate | 58138.0 | 58 | 3 | 8 | 10 | 4 | 7 | 0 | 1 | 69 | 663 | 1617 | 0 | Alone |
| **1** | Graduate | 46344.0 | 38 | 2 | 1 | 1 | 2 | 5 | 0 | 0 | 72 | 113 | 27 | 2 | Alone |
| **2** | Graduate | 71613.0 | 26 | 1 | 8 | 2 | 10 | 4 | 0 | 0 | 61 | 312 | 776 | 0 | Partner |
| **3** | Graduate | 26646.0 | 26 | 2 | 2 | 0 | 4 | 6 | 0 | 0 | 42 | 139 | 53 | 1 | Partner |
| **4** | Postgraduate | 58293.0 | 94 | 5 | 5 | 3 | 6 | 5 | 0 | 0 | 45 | 161 | 422 | 1 | Partner |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| **2235** | Graduate | 61223.0 | 46 | 2 | 9 | 3 | 4 | 5 | 0 | 0 | 59 | 381 | 1341 | 1 | Partner |
| **2236** | Postgraduate | 64014.0 | 56 | 7 | 8 | 2 | 5 | 7 | 0 | 0 | 80 | 19 | 444 | 3 | Partner |
| **2237** | Graduate | 56981.0 | 91 | 1 | 2 | 3 | 13 | 6 | 0 | 0 | 45 | 155 | 1241 | 0 | Alone |
| **2238** | Postgraduate | 69245.0 | 8 | 2 | 6 | 5 | 10 | 3 | 0 | 0 | 70 | 156 | 843 | 1 | Partner |
| **2239** | Postgraduate | 52869.0 | 40 | 3 | 3 | 1 | 4 | 7 | 0 | 1 | 72 | 622 | 172 | 2 | Partner |

2240 rows × 15 columns

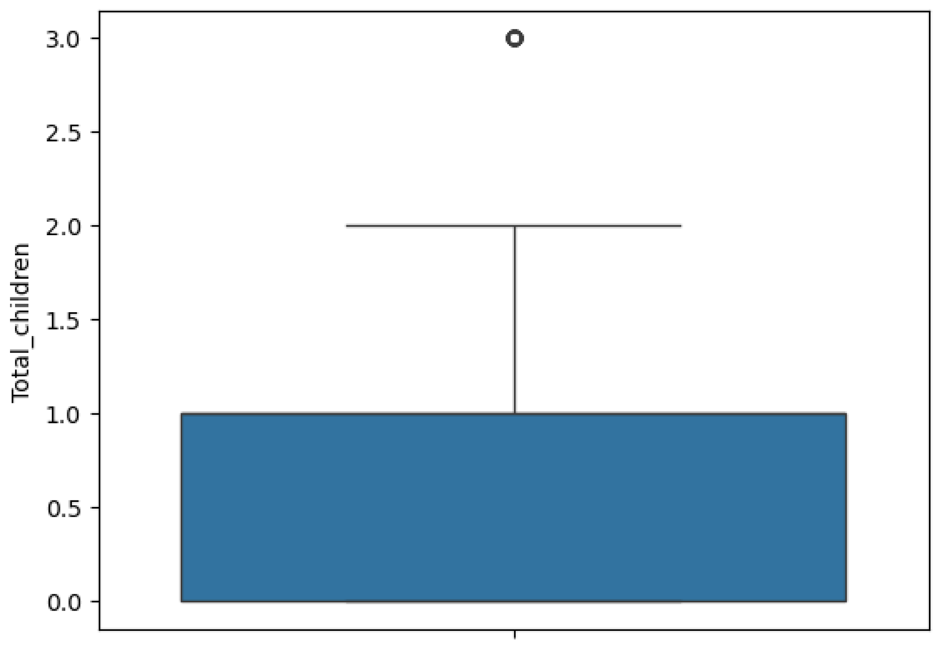
In [39]:
```
#Outliers
sns.boxplot(data=df_cleaned,y="Age")
```

Out[39]: <Axes: ylabel='Age'>



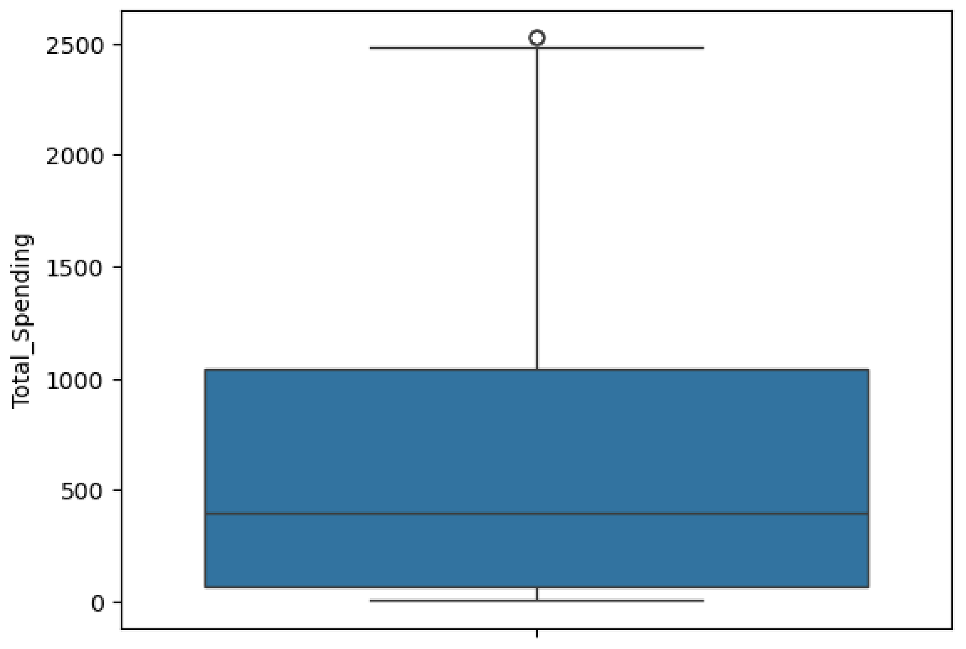

In [40]:
```
sns.boxplot(data=df,y="Total_children")
```

Out[40]: <Axes: ylabel='Total_children'>



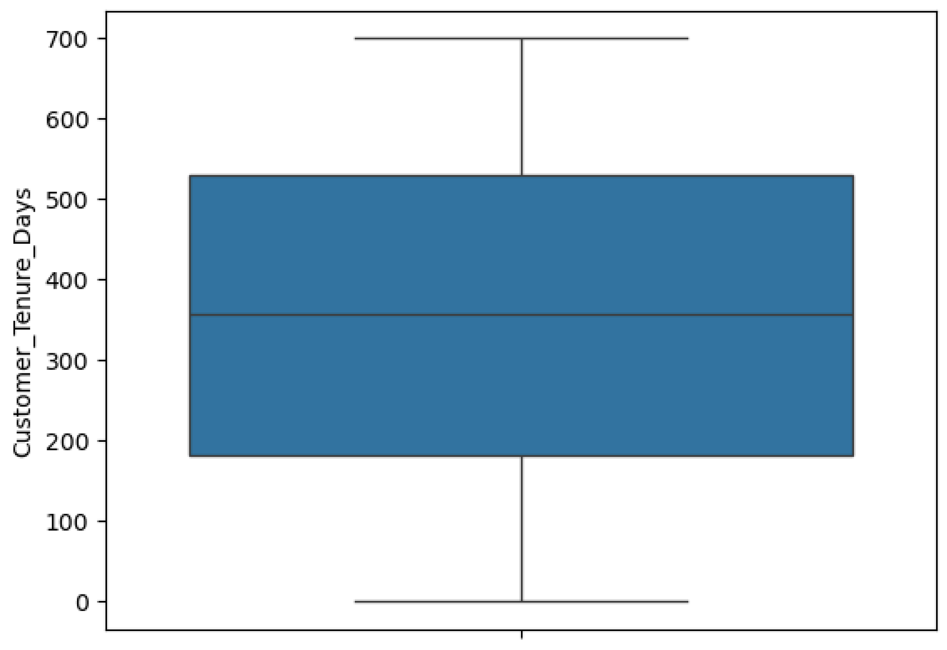

In [41]:
```
sns.boxplot(data=df_cleaned,y="Total_Spending")
```

Out[41]: <Axes: ylabel='Total_Spending'>



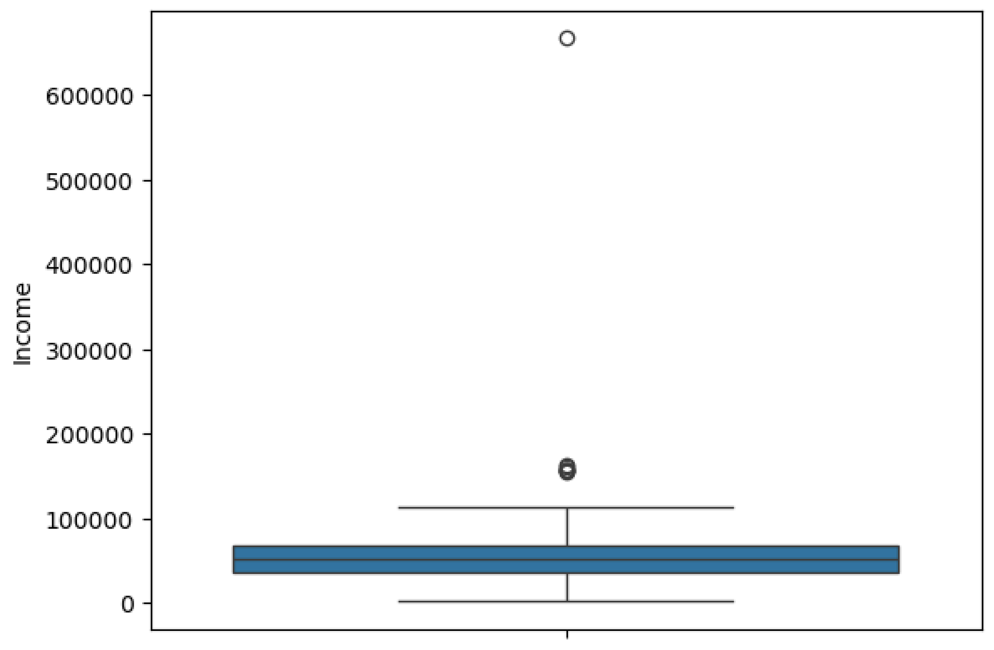

In [42]:
```
sns.boxplot(data=df_cleaned,y="Customer_Tenure_Days")
```

Out[42]: <Axes: ylabel='Customer_Tenure_Days'>




In [43]:
```
sns.boxplot(data=df_cleaned,y="Income")
```

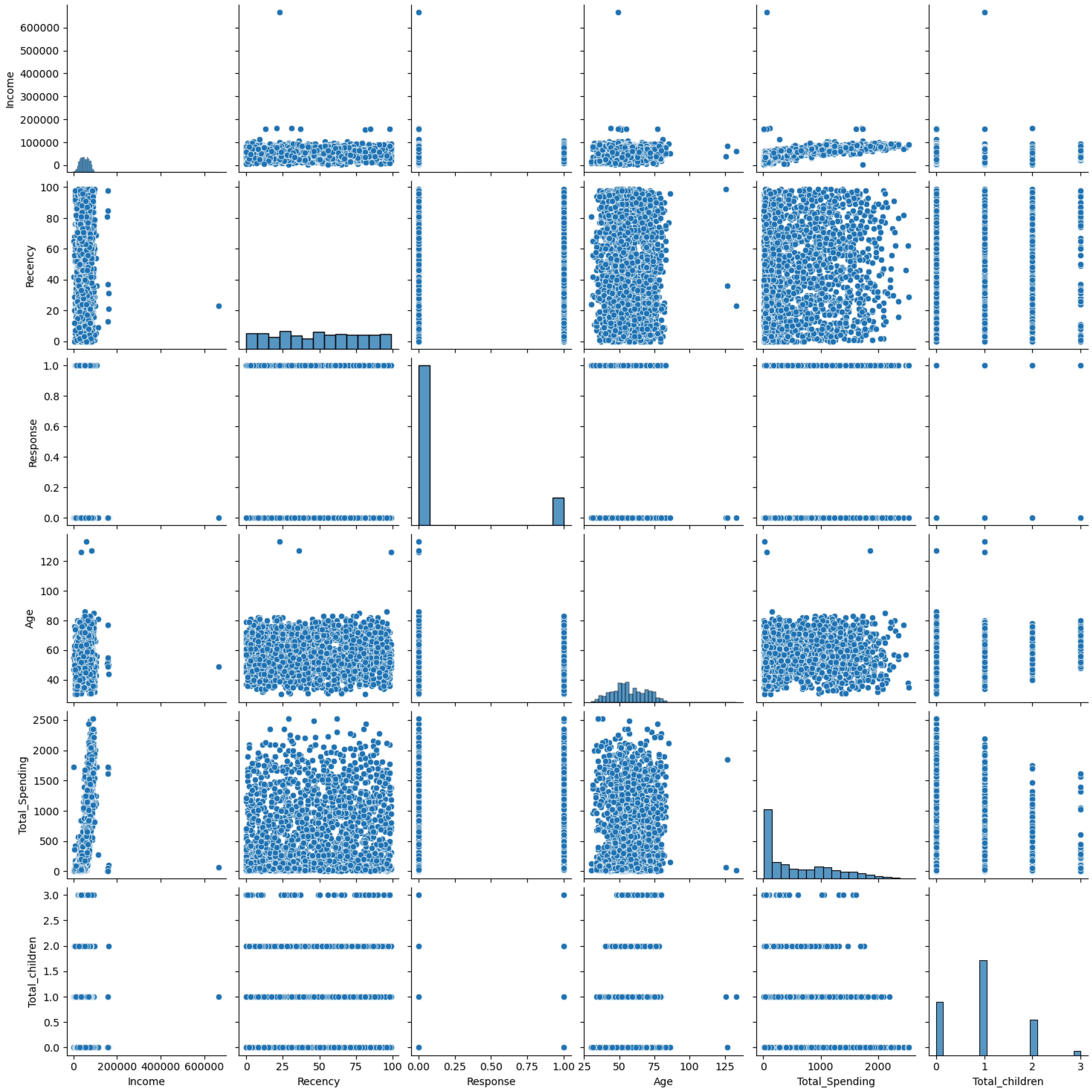
Out[43]: <Axes: ylabel='Income'>




In [44]:
```
cols=["Income","Recency","Response","Age","Total_Spending","Total_children"]
```

In [45]:
```
sns.pairplot(df_cleaned[cols])
```

Out[45]: <seaborn.axisgrid.PairGrid at 0x235767316a0>




In [46]:
```python
print("Data Size without outlier removal:",len(df_cleaned))
```

```
Data Size without outlier removal: 2240
```

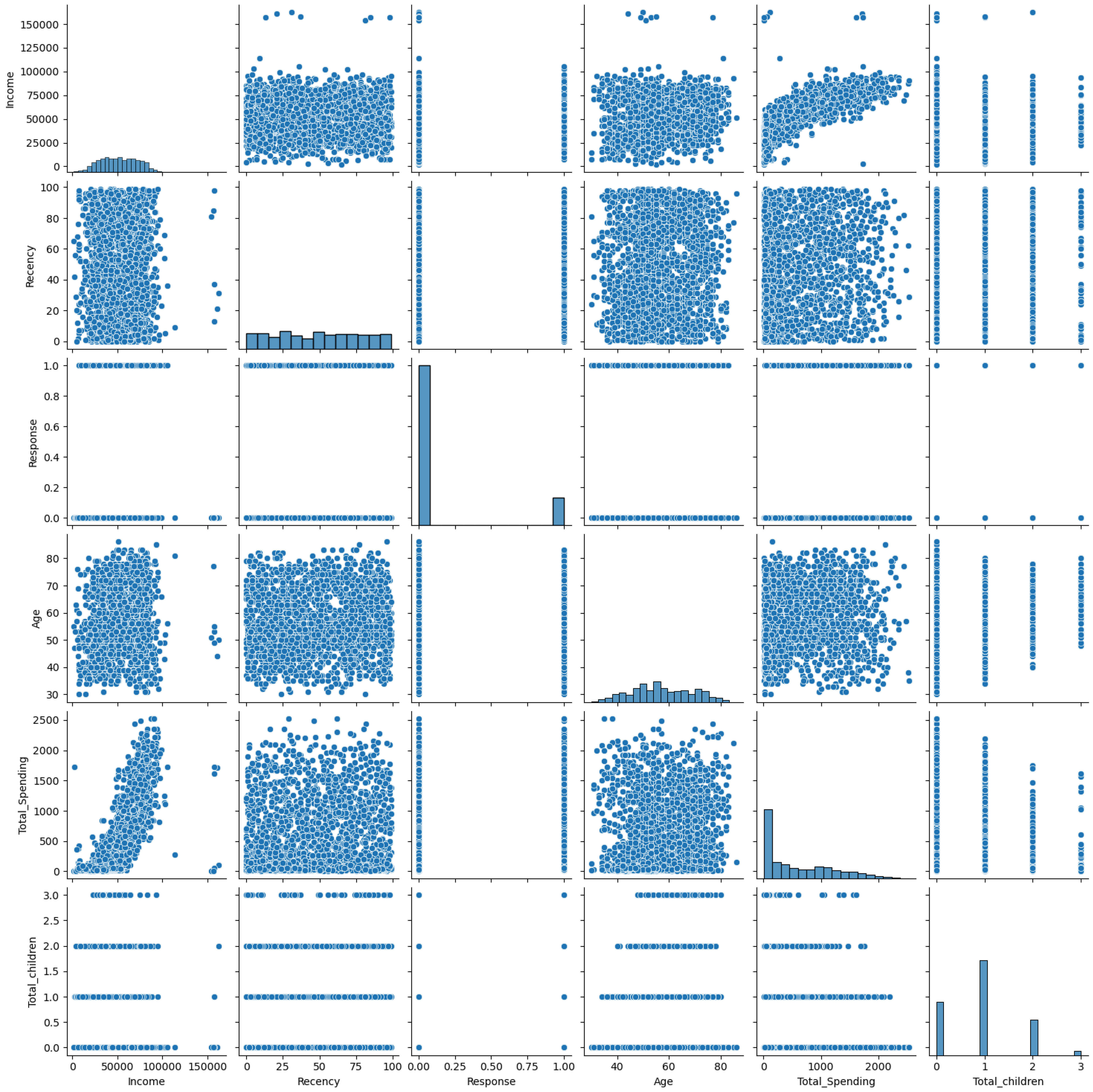
In [47]:
```python
#Remove outliers
df_cleaned=df_cleaned[(df_cleaned["Age"]<90)]
df_cleaned=df_cleaned[(df_cleaned["Income"]<600000)]
```

In [48]:
```python
print("Data Size with outlier removal:",len(df_cleaned))
```

```
Data Size with outlier removal: 2236
```

In [49]:
```python
sns.pairplot(df_cleaned[cols])
```

Out[49]:
```
<seaborn.axisgrid.PairGrid at 0x2357a9be490>
```




In [50]:
```
num_cols=df_cleaned.select_dtypes(include="number")
```

In [51]:
```
num_cols
```

Out[51]:

| | Income | Recency | NumDealsPurchases | NumWebPurchases | NumCatalogPurchases | NumStorePurchases | NumWebVisitsMonth | Complain | Response | Age | Customer_Tenure_Days | Total_Spending | Total_children |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 58138.0 | 58 | 3 | 8 | 10 | 4 | 7 | 0 | 1 | 69 | 663 | 1617 | 0 |
| **1** | 46344.0 | 38 | 2 | 1 | 1 | 2 | 5 | 0 | 0 | 72 | 113 | 27 | 2 |
| **2** | 71613.0 | 26 | 1 | 8 | 2 | 10 | 4 | 0 | 0 | 61 | 312 | 776 | 0 |
| **3** | 26646.0 | 26 | 2 | 2 | 0 | 4 | 6 | 0 | 0 | 42 | 139 | 53 | 1 |
| **4** | 58293.0 | 94 | 5 | 5 | 3 | 6 | 5 | 0 | 0 | 45 | 161 | 422 | 1 |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| **2235** | 61223.0 | 46 | 2 | 9 | 3 | 4 | 5 | 0 | 0 | 59 | 381 | 1341 | 1 |
| **2236** | 64014.0 | 56 | 7 | 8 | 2 | 5 | 7 | 0 | 0 | 80 | 19 | 444 | 3 |
| **2237** | 56981.0 | 91 | 1 | 2 | 3 | 13 | 6 | 0 | 0 | 45 | 155 | 1241 | 0 |
| **2238** | 69245.0 | 8 | 2 | 6 | 5 | 10 | 3 | 0 | 0 | 70 | 156 | 843 | 1 |
| **2239** | 52869.0 | 40 | 3 | 3 | 1 | 4 | 7 | 0 | 1 | 72 | 622 | 172 | 2 |

2236 rows × 13 columns

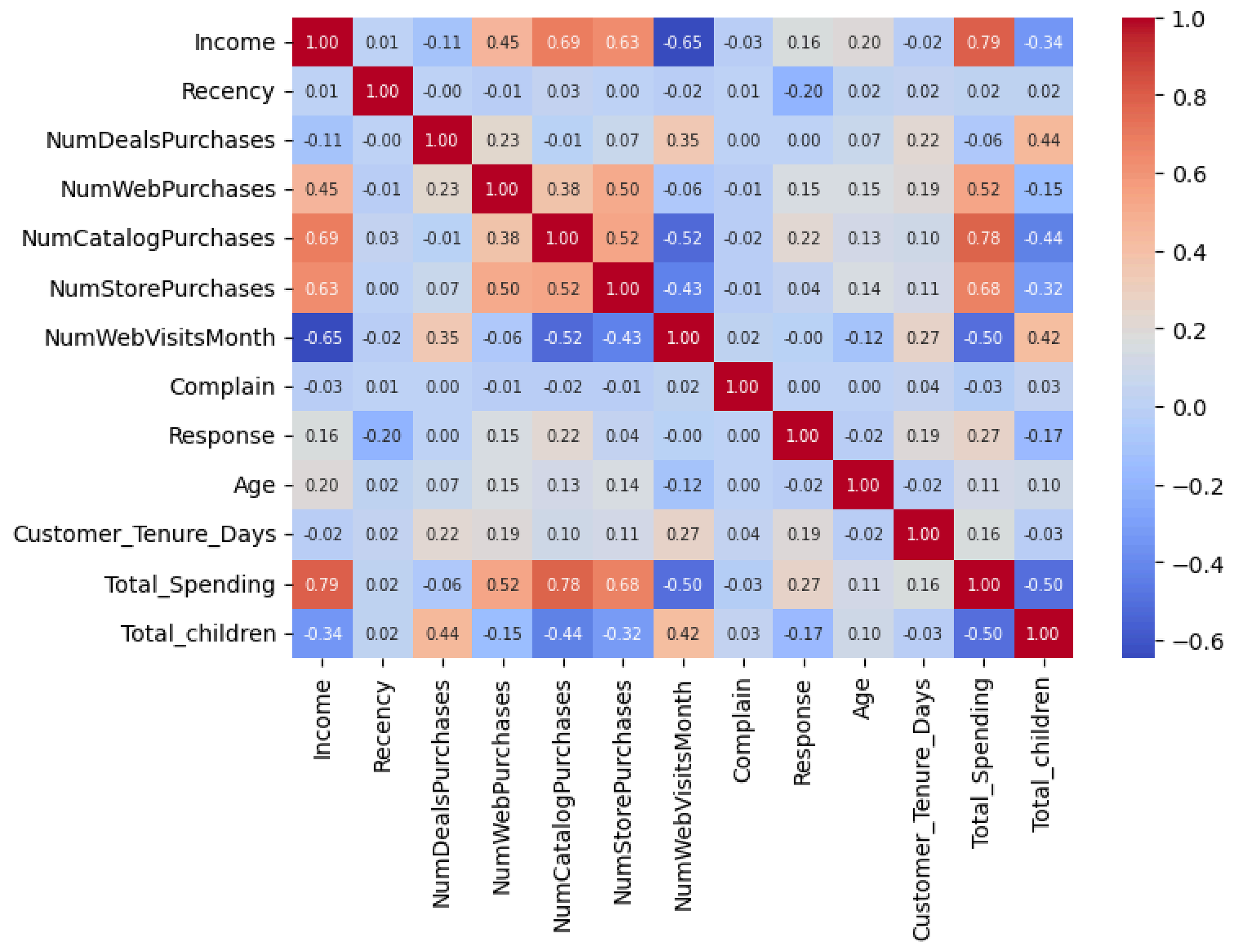
In [52]:
```
corr_matrix=num_cols.corr()
```

In [53]:
```
corr_matrix
```

Out[53]:

| | Income | Recency | NumDealsPurchases | NumWebPurchases | NumCatalogPurchases | NumStorePurchases | NumWebVisitsMonth | Complain | Response | Age | Customer_Tenure_Days | Total_Spending | Total_children |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Income** | 1.000000 | 0.007829 | -0.107169 | 0.450584 | 0.693781 | 0.628075 | -0.646382 | -0.027871 | 0.161121 | 0.198835 | -0.023677 | 0.789375 | -0.340550 |
| **Recency** | 0.007829 | 1.000000 | -0.000638 | -0.010776 | 0.025226 | 0.000771 | -0.021850 | 0.005361 | -0.198781 | 0.019396 | 0.024238 | 0.020842 | 0.017826 |
| **NumDealsPurchases** | -0.107169 | -0.000638 | 1.000000 | 0.233971 | -0.008510 | 0.068418 | 0.347216 | 0.003636 | 0.002017 | 0.068286 | 0.218009 | -0.064831 | 0.439682 |
| **NumWebPurchases** | 0.450584 | -0.010776 | 0.233971 | 1.000000 | 0.378049 | 0.502227 | -0.056204 | -0.013250 | 0.148390 | 0.153873 | 0.191211 | 0.519948 | -0.146429 |
| **NumCatalogPurchases** | 0.693781 | 0.025226 | -0.008510 | 0.378049 | 1.000000 | 0.518788 | -0.520376 | -0.018304 | 0.220813 | 0.125285 | 0.095836 | 0.778343 | -0.439631 |
| **NumStorePurchases** | 0.628075 | 0.000771 | 0.068418 | 0.502227 | 0.518788 | 1.000000 | -0.429857 | -0.011563 | 0.038702 | 0.139237 | 0.109727 | 0.675460 | -0.321729 |
| **NumWebVisitsMonth** | -0.646382 | -0.021850 | 0.347216 | -0.056204 | -0.520376 | -0.429857 | 1.000000 | 0.020796 | -0.004397 | -0.117498 | 0.272105 | -0.499909 | 0.417908 |
| **Complain** | -0.027871 | 0.005361 | 0.003636 | -0.013250 | -0.018304 | -0.011563 | 0.020796 | 1.000000 | 0.000167 | 0.004450 | 0.035685 | -0.033784 | 0.031480 |
| **Response** | 0.161121 | -0.198781 | 0.002017 | 0.148390 | 0.220813 | 0.038702 | -0.004397 | 0.000167 | 1.000000 | -0.018557 | 0.194232 | 0.265615 | -0.169451 |
| **Age** | 0.198835 | 0.019396 | 0.068286 | 0.153873 | 0.125285 | 0.139237 | -0.117498 | 0.004450 | -0.018557 | 1.000000 | -0.016451 | 0.113618 | 0.095512 |
| **Customer_Tenure_Days** | -0.023677 | 0.024238 | 0.218009 | 0.191211 | 0.095836 | 0.109727 | 0.272105 | 0.035685 | 0.194232 | -0.016451 | 1.000000 | 0.158764 | -0.025742 |
| **Total_Spending** | 0.789375 | 0.020842 | -0.064831 | 0.519948 | 0.778343 | 0.675460 | -0.499909 | -0.033784 | 0.265615 | 0.113618 | 0.158764 | 1.000000 | -0.498579 |
| **Total_children** | -0.340550 | 0.017826 | 0.439682 | -0.146429 | -0.439631 | -0.321729 | 0.417908 | 0.031480 | -0.169451 | 0.095512 | -0.025742 | -0.498579 | 1.000000 |

In [54]:
```
plt.figure(figsize=(8,6))
sns.heatmap(corr_matrix,annot=True,fmt=".2f",cmap="coolwarm",annot_kws={"size":7})
plt.tight_layout()
```




## Feature Encoding

In [55]:
```
from sklearn.preprocessing import OneHotEncoder
```

In [56]:
```
ohe=OneHotEncoder()
cat_cols=["Education","Living_with"]
enc_cols=ohe.fit_transform(df_cleaned[cat_cols])
```

In [57]:
```
enc_df=pd.DataFrame(enc_cols.toarray(),columns=ohe.get_feature_names_out(cat_cols),index=df_cleaned.index)
```

In [58]:
```
df_encoded=pd.concat([df_cleaned.drop(columns=cat_cols),enc_df],axis=1)
```

In [59]:
```
df_encoded
```

Out[59]:

| | Income | Recency | NumDealsPurchases | NumWebPurchases | NumCatalogPurchases | NumStorePurchases | NumWebVisitsMonth | Complain | Response | Age | Customer_Tenure_Days | Total_Spending | Total_children | Education_Graduate | Education_Postgraduate | Education_Undergraduate | Living_with_Alone | Livin |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 58138.0 | 58 | 3 | 8 | 10 | 4 | 7 | 0 | 1 | 69 | 663 | 1617 | 0 | 1.0 | 0.0 | 0.0 | 1.0 | |
| **1** | 46344.0 | 38 | 2 | 1 | 1 | 2 | 5 | 0 | 0 | 72 | 113 | 27 | 2 | 1.0 | 0.0 | 0.0 | 1.0 | |
| **2** | 71613.0 | 26 | 1 | 8 | 2 | 10 | 4 | 0 | 0 | 61 | 312 | 776 | 0 | 1.0 | 0.0 | 0.0 | 0.0 | |
| **3** | 26646.0 | 26 | 2 | 2 | 0 | 4 | 6 | 0 | 0 | 42 | 139 | 53 | 1 | 1.0 | 0.0 | 0.0 | 0.0 | |
| **4** | 58293.0 | 94 | 5 | 5 | 3 | 6 | 5 | 0 | 0 | 45 | 161 | 422 | 1 | 0.0 | 1.0 | 0.0 | 0.0 | |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | |
| **2235** | 61223.0 | 46 | 2 | 9 | 3 | 4 | 5 | 0 | 0 | 59 | 381 | 1341 | 1 | 1.0 | 0.0 | 0.0 | 0.0 | |
| **2236** | 64014.0 | 56 | 7 | 8 | 2 | 5 | 7 | 0 | 0 | 80 | 19 | 444 | 3 | 0.0 | 1.0 | 0.0 | 0.0 | |
| **2237** | 56981.0 | 91 | 1 | 2 | 3 | 13 | 6 | 0 | 0 | 45 | 155 | 1241 | 0 | 1.0 | 0.0 | 0.0 | 1.0 | |
| **2238** | 69245.0 | 8 | 2 | 6 | 5 | 10 | 3 | 0 | 0 | 70 | 156 | 843 | 1 | 0.0 | 1.0 | 0.0 | 0.0 | |
| **2239** | 52869.0 | 40 | 3 | 3 | 1 | 4 | 7 | 0 | 1 | 72 | 622 | 172 | 2 | 0.0 | 1.0 | 0.0 | 0.0 | |

2236 rows × 18 columns

In [60]:
```
from sklearn.preprocessing import StandardScaler
scaler=StandardScaler()
df_scaled=scaler.fit_transform(df_encoded)
```

In [61]:
```
df_scaled.shape
```

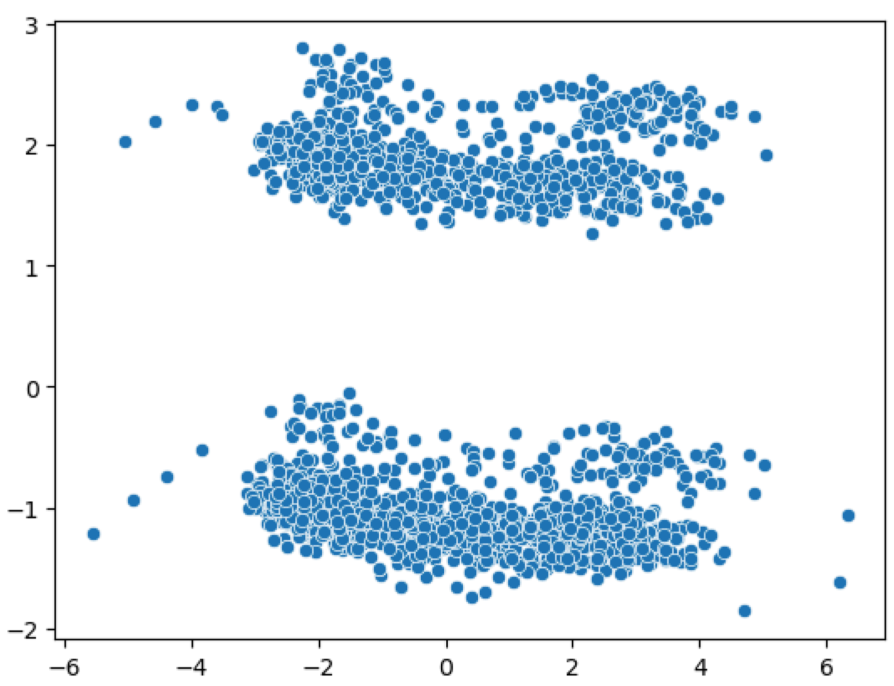
Out[61]: (2236, 18)

In [62]:
```
#2D Visulization
from sklearn.decomposition import PCA
pca=PCA(n_components=2,random_state=42)
```

In [63]:
```
X_pca=pca.fit_transform(df_scaled)
```

In [64]:
```
sns.scatterplot(x=X_pca[:,0],y=X_pca[:,1])
```

Out[64]: <Axes: >




In [65]:
```
pca.explained_variance_ratio_
```

Out[65]: array([0.23163158, 0.11385454])

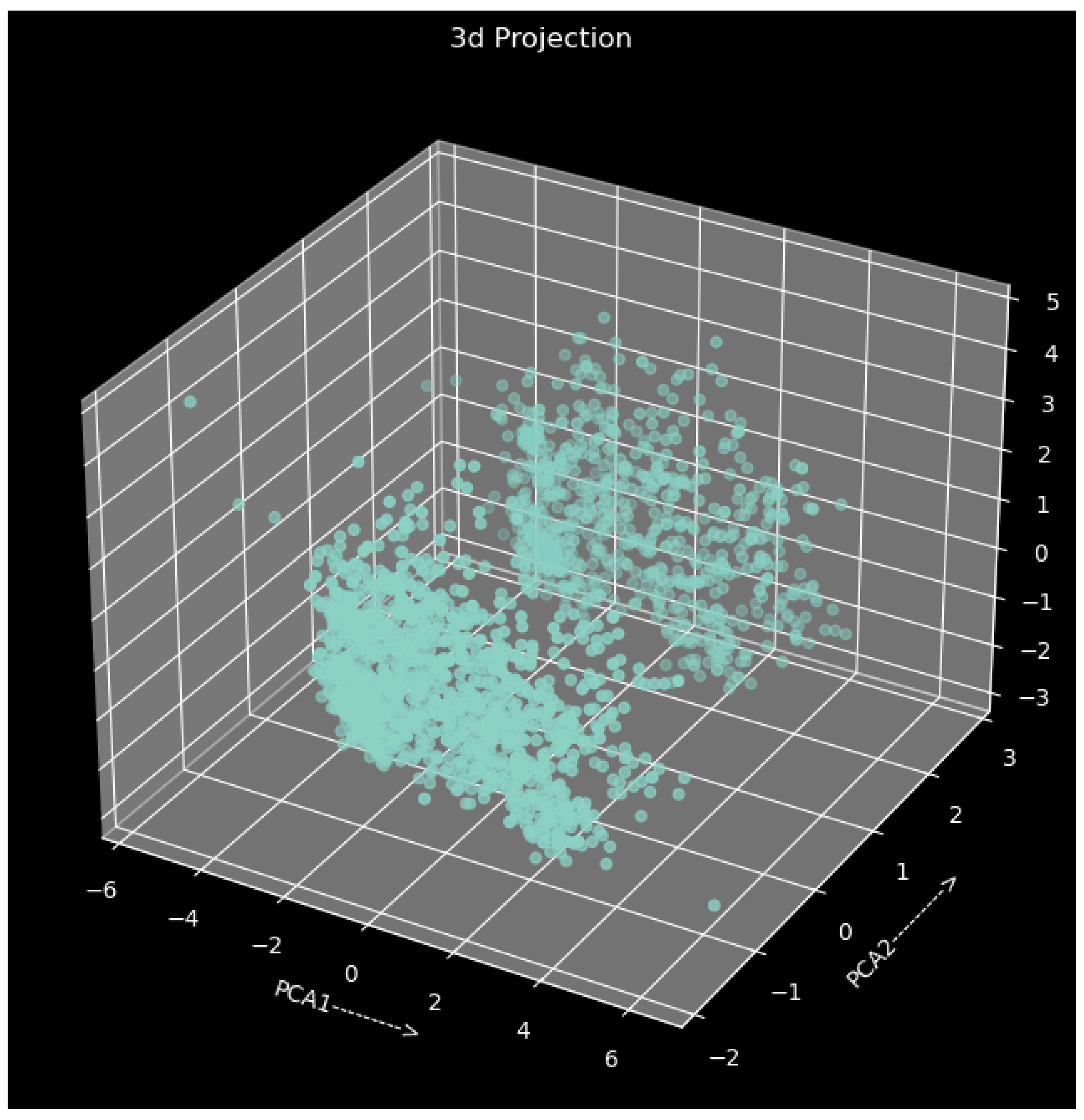
In [66]:
```
pca2=PCA(n_components=3,random_state=42)
```

In [67]:
```
X_pca2=pca2.fit_transform(df_scaled)
pca2.explained_variance_ratio_
```

Out[67]: array([0.23163158, 0.11385454, 0.10405815])

In [68]:
```
#Plot
plt.style.use("dark_background")
fig=plt.figure(figsize=(10,8))
ax=fig.add_subplot(111,projection="3d")
ax.scatter(X_pca2[:,0],X_pca2[:,1],X_pca2[:,2])
ax.set_xlabel("PCA1--------->")
ax.set_ylabel("PCA2--------->")
ax.set_zlabel("PCA3--------->")
ax.set_title("3d Projection")
```

Out[68]: Text(0.5, 0.92, '3d Projection')

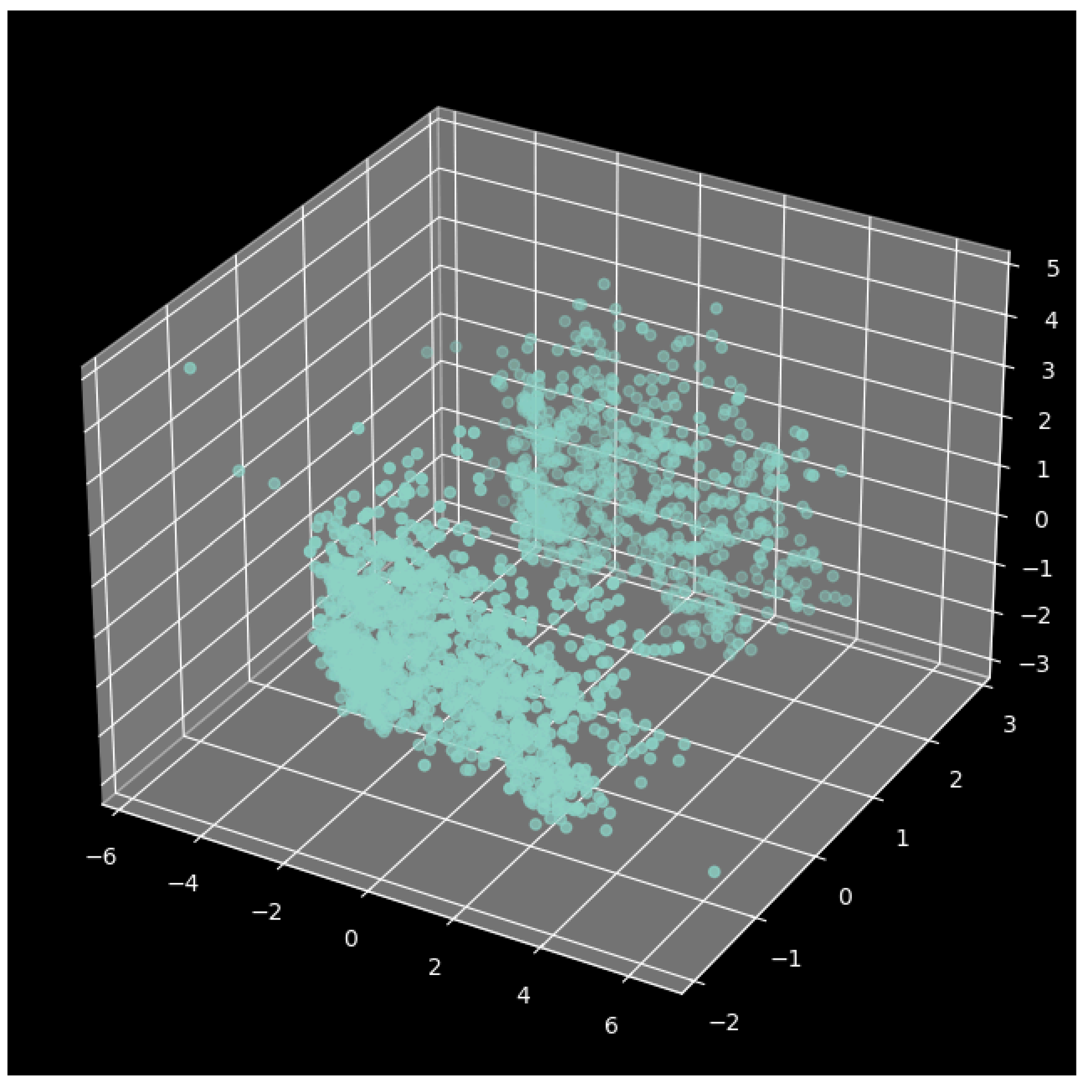



In [69]:
```
plt.style.available
```

Out[69]:
```
['Solarize_Light2',
 '_classic_test_patch',
 '_mpl-gallery',
 '_mpl-gallery-nogrid',
 'bmh',
 'classic',
 'dark_background',
 'fast',
 'fivethirtyeight',
 'ggplot',
 'grayscale',
 'petroff10',
 'seaborn-v0_8',
 'seaborn-v0_8-bright',
 'seaborn-v0_8-colorblind',
 'seaborn-v0_8-dark',
 'seaborn-v0_8-dark-palette',
 'seaborn-v0_8-darkgrid',
 'seaborn-v0_8-deep',
 'seaborn-v0_8-muted',
 'seaborn-v0_8-notebook',
 'seaborn-v0_8-paper',
 'seaborn-v0_8-pastel',
 'seaborn-v0_8-poster',
 'seaborn-v0_8-talk',
 'seaborn-v0_8-ticks',
 'seaborn-v0_8-white',
 'seaborn-v0_8-whitegrid',
 'tableau-colorblind10']
```

In [70]:
```
fig=plt.figure(figsize=(10,8))
ax=fig.add_subplot(111,projection="3d")
ax.scatter(X_pca2[:,0],X_pca2[:,1],X_pca2[:,2])
```

Out[70]: <mpl_toolkits.mplot3d.art3d.Path3DCollection at 0x23501e8e210>



In [71]:
```
pca2.explained_variance_ratio_
```

Out[71]:
```
array([0.23163158, 0.11385454, 0.10405815])
```

# Analyze K value

## Elbow Method

In [72]:
```
X_pca2
```

Out[72]:
```
array([[ 2.7087298 ,  2.39869224,  0.19587886],
       [-1.76655338,  1.82575242, -0.85552989],
       [ 1.81995208, -1.07430148, -1.27485084],
       ...,
       [ 1.42793889,  1.81716429, -1.87941234],
       [ 1.90073104, -1.4254157 ,  1.31005531],
       [-0.96995283, -0.67250249,  2.56404824]])
```

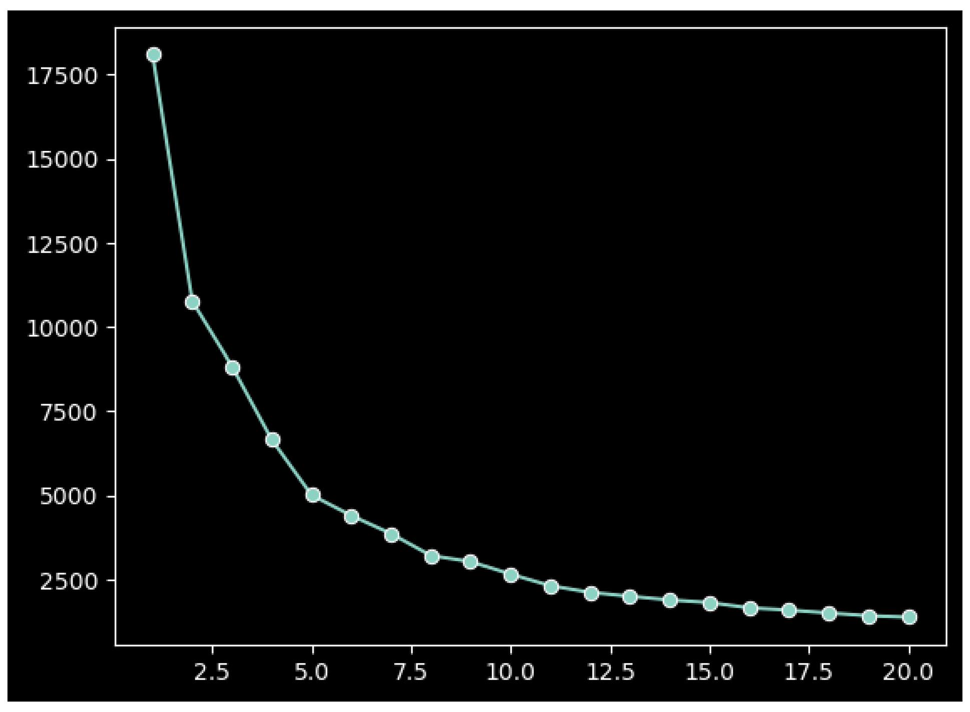
In [73]:
```
wcss=[]
from sklearn.cluster import KMeans
from kneed import KneeLocator
for k in range(1,21):
    kmeans=KMeans(n_clusters=k,random_state=42)
    kmeans.fit(X_pca2)
    wcss.append(kmeans.inertia_)
```

```
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
```

In [74]:
```
knee=KneeLocator(range(1,21),wcss,curve="convex",direction="decreasing")
print("Optimal K=",knee.knee)
sns.lineplot(x=range(1,21),y=wcss,marker="o")
```

```
Optimal K= 5
```

Out[74]: `<Axes: >`




## Silhouette Score

In [75]:
```
from sklearn.metrics import silhouette_score
scores=[]
for k in range(2,21):
    kmeans=KMeans(n_clusters=k,random_state=42)
    labels=kmeans.fit_predict(X_pca2)
    scores.append(silhouette_score(X_pca2,labels))
```

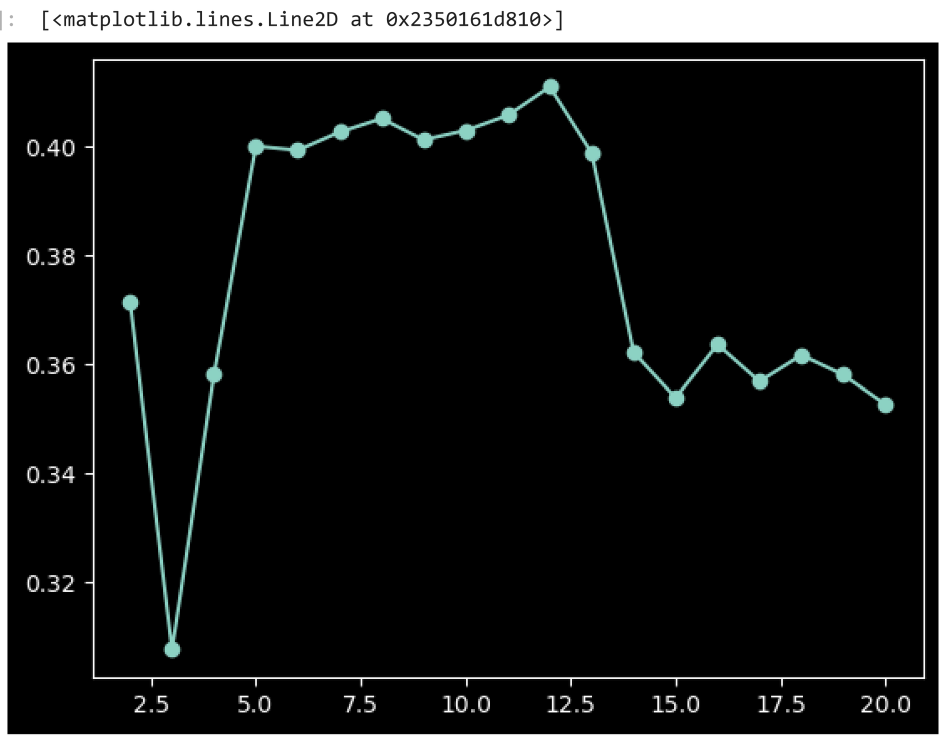
```
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
```

In [76]:
```
scores
```

Out[76]:
```
[np.float64(0.37157939079769425),
 np.float64(0.30768819274753195),
 np.float64(0.35809961595980194),
 np.float64(0.4000152147663037),
 np.float64(0.3992918389858723),
 np.float64(0.40262087405621194),
 np.float64(0.4050858828695181),
 np.float64(0.4011889704657738),
 np.float64(0.40291173364870503),
 np.float64(0.40568550196597225),
 np.float64(0.4109146649763734),
 np.float64(0.3987399154391176),
 np.float64(0.3622069397491765),
 np.float64(0.3538628001189791),
 np.float64(0.3636397332649718),
 np.float64(0.3568735131053515),
 np.float64(0.3616924344392367),
 np.float64(0.35808651986994045),
 np.float64(0.3526866644064818)]
```

In [77]:
```
plt.plot(range(2,21),scores,marker="o")
```

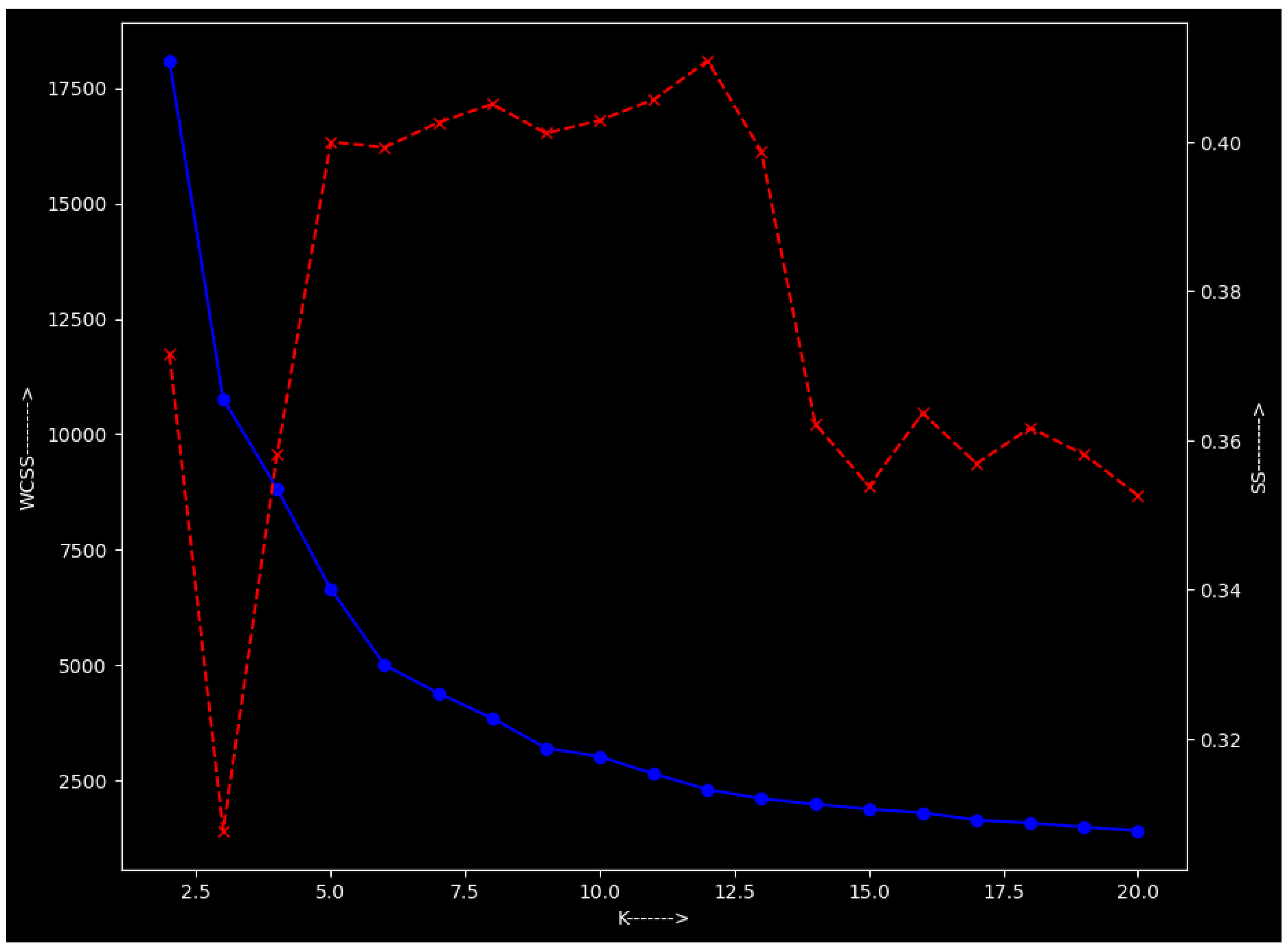
Out[77]: `[<matplotlib.lines.Line2D at 0x2350161d810>]`




In [78]:
```
#Combined Plot
k_range=range(2,21)
fig,ax1=plt.subplots(figsize=(10,8))
ax1.plot(k_range,wcss[:len(k_range)],marker="o",color="blue")
ax1.set_xlabel("K------->")
ax1.set_ylabel("WCSS-------->")
ax2=ax1.twinx()
```

```
ax2.plot(k_range,scores[:len(k_range)],marker="x",color="red",linestyle="--")
ax2.set_ylabel("SS-------->")
```

Out[78]: Text(0, 0.5, 'SS-------->')




In [79]:
```
import numpy as np
print(type(wcss))
print(np.array(wcss).shape)
```

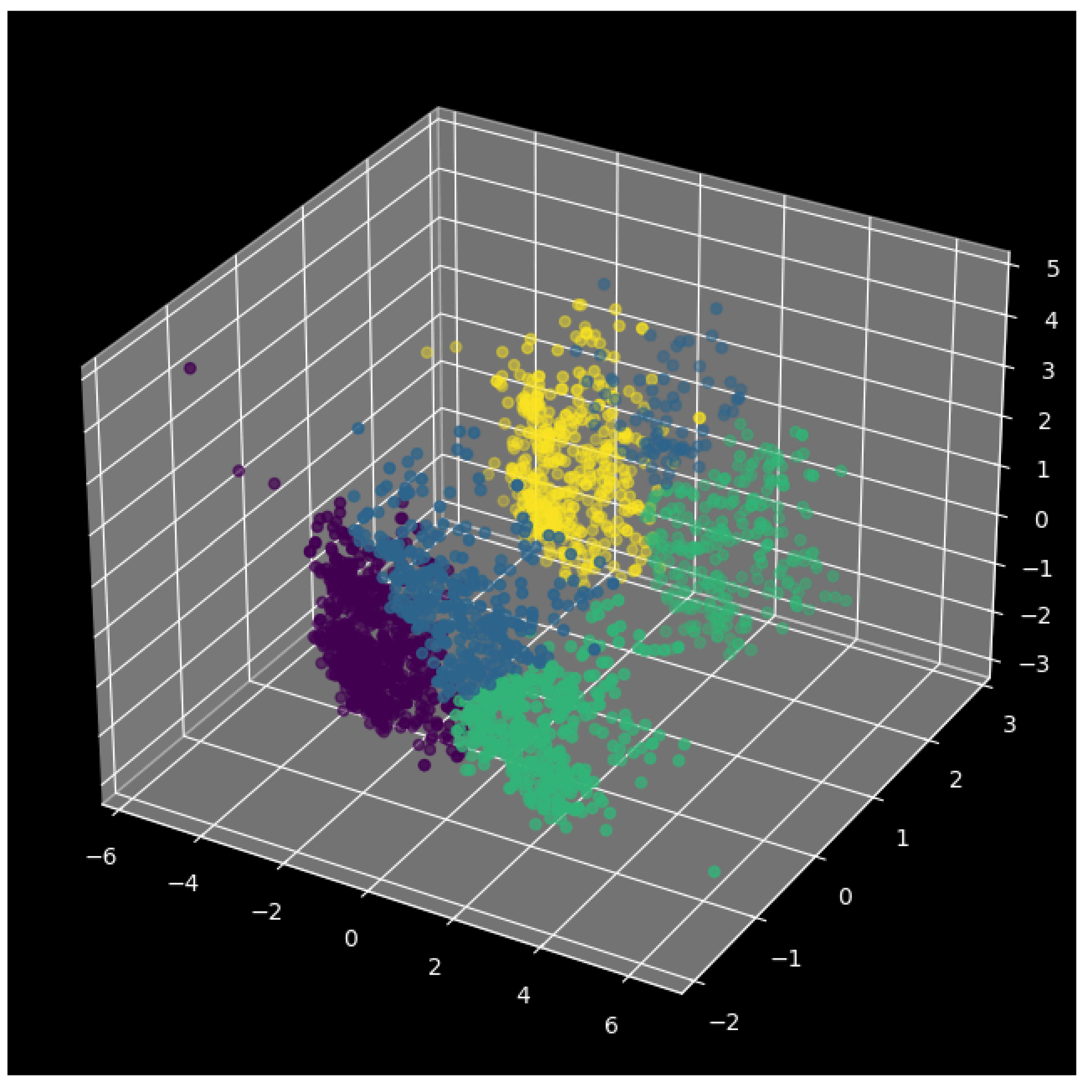
```
<class 'list'>
(20,)
```

In [80]:
```
kmeans=KMeans(n_clusters=4,random_state=42)
labels_kmeans=kmeans.fit_predict(X_pca2)
```

```
C:\ProgramData\Anaconda\Lib\site-packages\sklearn\cluster\_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=9.
  warnings.warn(
```

In [81]:
```
fig=plt.figure(figsize=(10,8))
ax=fig.add_subplot(111,projection="3d")
ax.scatter(X_pca2[:,0],X_pca2[:,1],X_pca2[:,2],c=labels_kmeans)
```

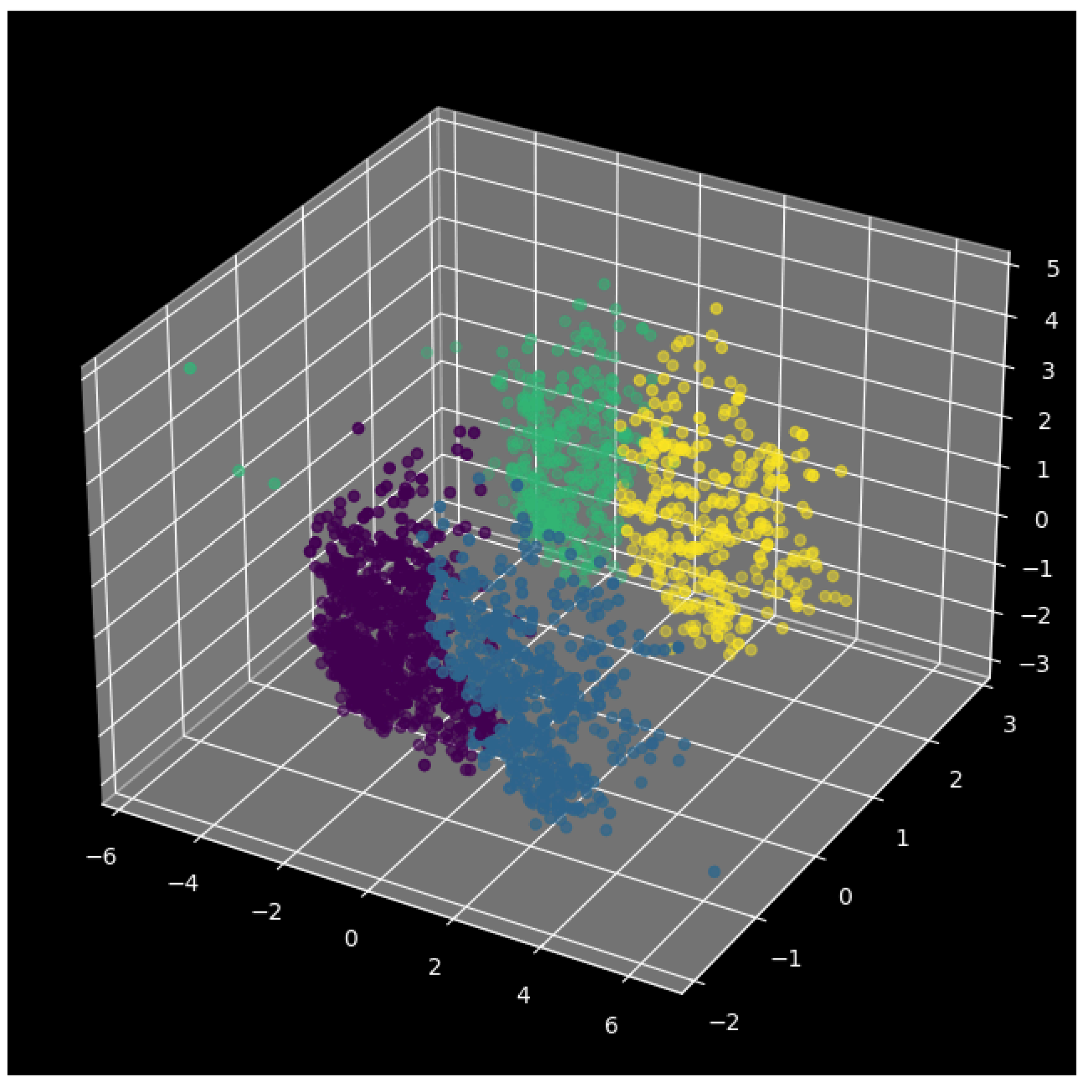
Out[81]: <mpl_toolkits.mplot3d.art3d.Path3DCollection at 0x23501766fd0>



In [82]:
```
from sklearn.cluster import AgglomerativeClustering
agg=AgglomerativeClustering(n_clusters=4,linkage='ward')
labels_agg=agg.fit_predict(X_pca2)
```

In [83]:
```
fig=plt.figure(figsize=(10,8))
ax=fig.add_subplot(111,projection="3d")
ax.scatter(X_pca2[:,0],X_pca2[:,1],X_pca2[:,2],c=labels_agg)
```

Out[83]: <mpl_toolkits.mplot3d.art3d.Path3DCollection at 0x2350296fed0>



## Characterization of Clusters

In [84]:
```
df_encoded["clusters"]=labels_agg
```

In [85]:
```
df_encoded
```

Out[85]:

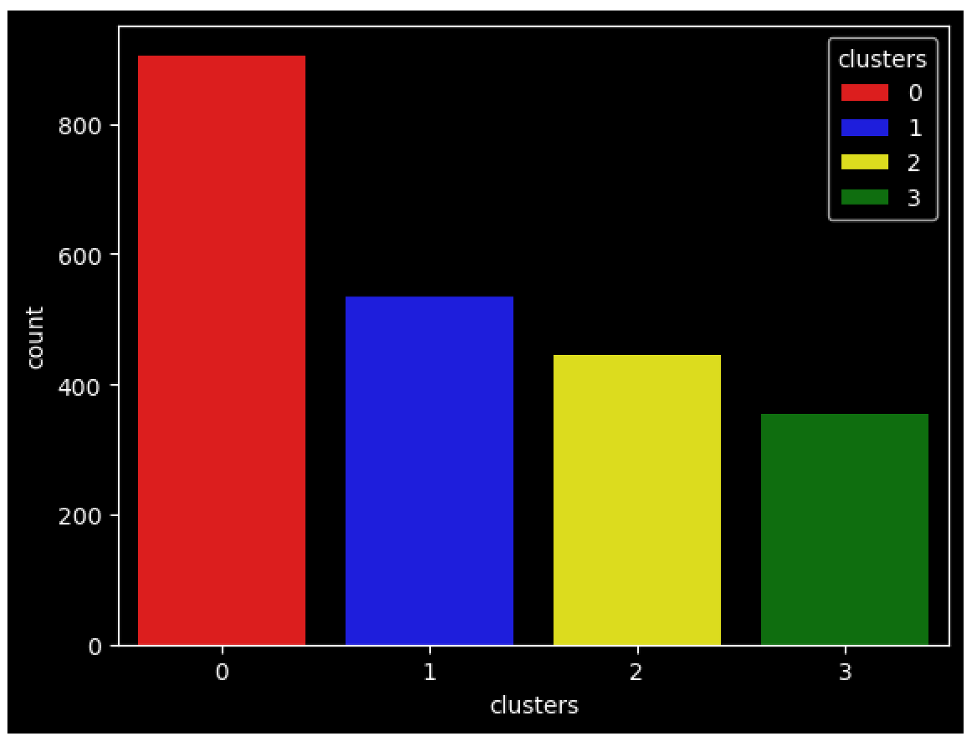
| | Income | Recency | NumDealsPurchases | NumWebPurchases | NumCatalogPurchases | NumStorePurchases | NumWebVisitsMonth | Complain | Response | Age | Customer_Tenure_Days | Total_Spending | Total_children | Education_Graduate | Education_Postgraduate | Education_Undergraduate | Living_with_Alone | Living |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 58138.0 | 58 | 3 | 8 | 10 | 4 | 7 | 0 | 1 | 69 | 663 | 1617 | 0 | 1.0 | 0.0 | 0.0 | 1.0 | |
| **1** | 46344.0 | 38 | 2 | 1 | 1 | 2 | 5 | 0 | 0 | 72 | 113 | 27 | 2 | 1.0 | 0.0 | 0.0 | 1.0 | |
| **2** | 71613.0 | 26 | 1 | 8 | 2 | 10 | 4 | 0 | 0 | 61 | 312 | 776 | 0 | 1.0 | 0.0 | 0.0 | 0.0 | |
| **3** | 26646.0 | 26 | 2 | 2 | 0 | 4 | 6 | 0 | 0 | 42 | 139 | 53 | 1 | 1.0 | 0.0 | 0.0 | 0.0 | |
| **4** | 58293.0 | 94 | 5 | 5 | 3 | 6 | 5 | 0 | 0 | 45 | 161 | 422 | 1 | 0.0 | 1.0 | 0.0 | 0.0 | |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | |
| **2235** | 61223.0 | 46 | 2 | 9 | 3 | 4 | 5 | 0 | 0 | 59 | 381 | 1341 | 1 | 1.0 | 0.0 | 0.0 | 0.0 | |
| **2236** | 64014.0 | 56 | 7 | 8 | 2 | 5 | 7 | 0 | 0 | 80 | 19 | 444 | 3 | 0.0 | 1.0 | 0.0 | 0.0 | |
| **2237** | 56981.0 | 91 | 1 | 2 | 3 | 13 | 6 | 0 | 0 | 45 | 155 | 1241 | 0 | 1.0 | 0.0 | 0.0 | 1.0 | |
| **2238** | 69245.0 | 8 | 2 | 6 | 5 | 10 | 3 | 0 | 0 | 70 | 156 | 843 | 1 | 0.0 | 1.0 | 0.0 | 0.0 | |
| **2239** | 52869.0 | 40 | 3 | 3 | 1 | 4 | 7 | 0 | 1 | 72 | 622 | 172 | 2 | 0.0 | 1.0 | 0.0 | 0.0 | |

2236 rows × 19 columns

In [86]:

```
pal=["red","blue","yellow","green"]
sns.countplot(x=df_encoded["clusters"],palette=pal,hue=df_encoded["clusters"])
```

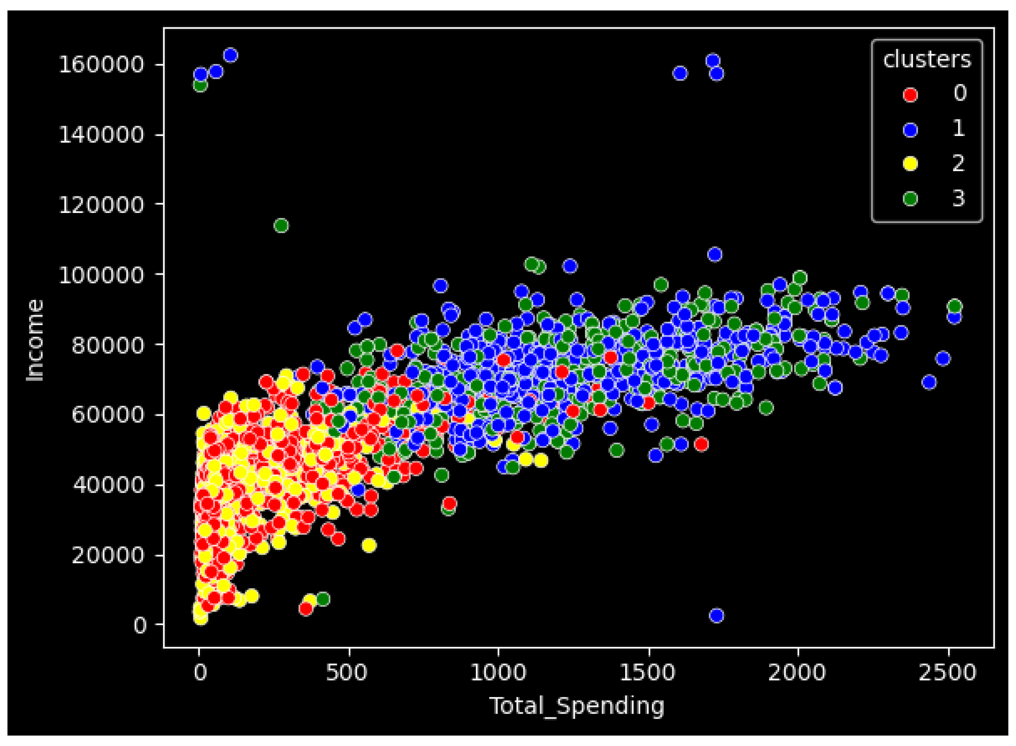
Out[86]: <Axes: xlabel='clusters', ylabel='count'>




In [87]:

```
#Income & Spending Patterns
sns.scatterplot(x=df_encoded["Total_Spending"],y=df_encoded["Income"],hue=df_encoded["clusters"],palette=pal)
```

Out[87]: <Axes: xlabel='Total_Spending', ylabel='Income'>




In [88]:

```
#Cluster Summary
cluster_summary=df_encoded.groupby("clusters").mean()
print(cluster_summary)
```

```
                Income    Recency  NumDealsPurchases  NumWebPurchases  \
clusters
0         39680.580110  48.914917           2.594475         3.153591
1         72808.445693  49.202247           1.958801         5.687266
2         36960.143018  48.319820           2.594595         2.713964
3         70722.681303  50.504249           1.855524         5.790368

          NumCatalogPurchases  NumStorePurchases  NumWebVisitsMonth  Complain  \
clusters
0                    0.969061           4.143646           6.307182  0.011050
1                    5.498127           8.659176           3.580524  0.005618
2                    0.837838           3.623874           6.659910  0.011261
3                    5.014164           8.430595           3.728045  0.005666

          Response        Age  Customer_Tenure_Days  Total_Spending  \
clusters
0         0.076243  55.669613            342.939227      221.955801
1         0.166667  59.492509            369.720974     1236.588015
2         0.141892  55.691441            338.781532      165.702703
3         0.320113  58.932011            376.280453     1190.385269

          Total_children  Education_Graduate  Education_Postgraduate  \
clusters
0               1.243094            0.514917                0.338122
1               0.511236            0.471910                0.455056
2               1.272523            0.488739                0.378378
3               0.461756            0.541076                0.390935

          Education_Undergraduate  Living_with_Alone  Living_with_Partner
clusters
0                        0.146961           0.000000             1.000000
1                        0.073034           0.000000             1.000000
2                        0.132883           0.993243             0.006757
3                        0.067989           1.000000             0.000000
```